SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**RAVURU PRANATHI**
Register Number	**192525076**

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)
Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 4

Duration : 60 Minutes

Code of Conduct
I R.Pranathi(192525076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: R. Pranathi

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM: To Create a simple cloud software application for Library Book Reservation System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

step 1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step 4: Enter application name as Library Book Reservation System

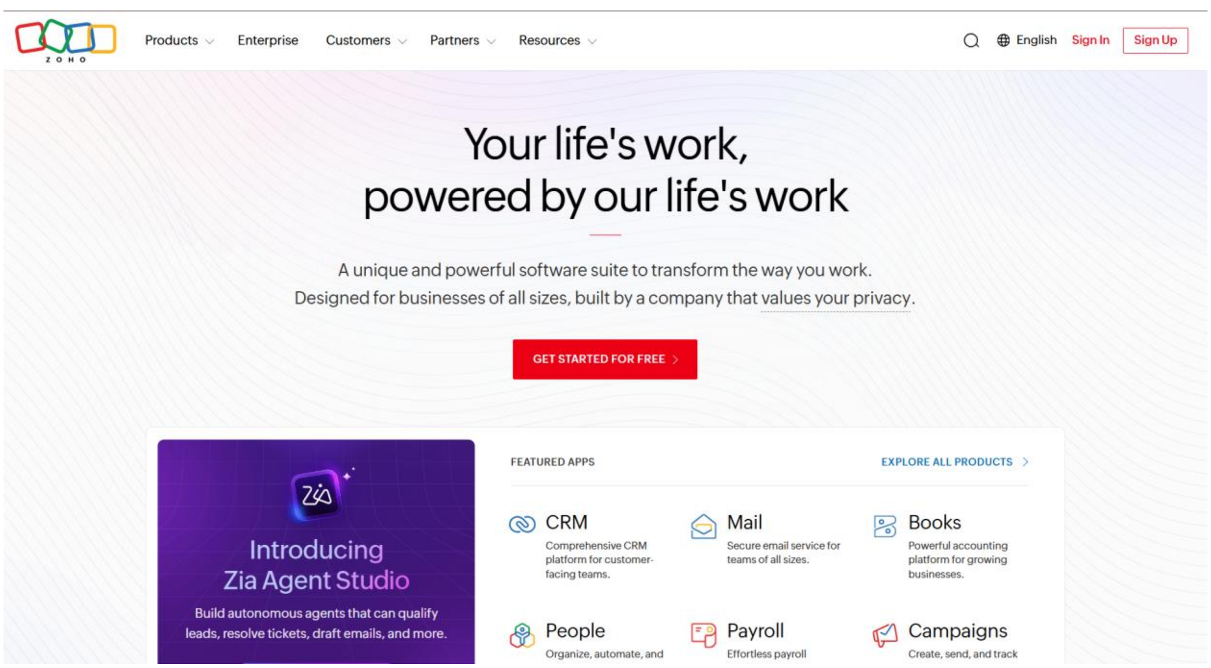
step 5: Created new application as Library Book Reservation System

step 6: Select one form

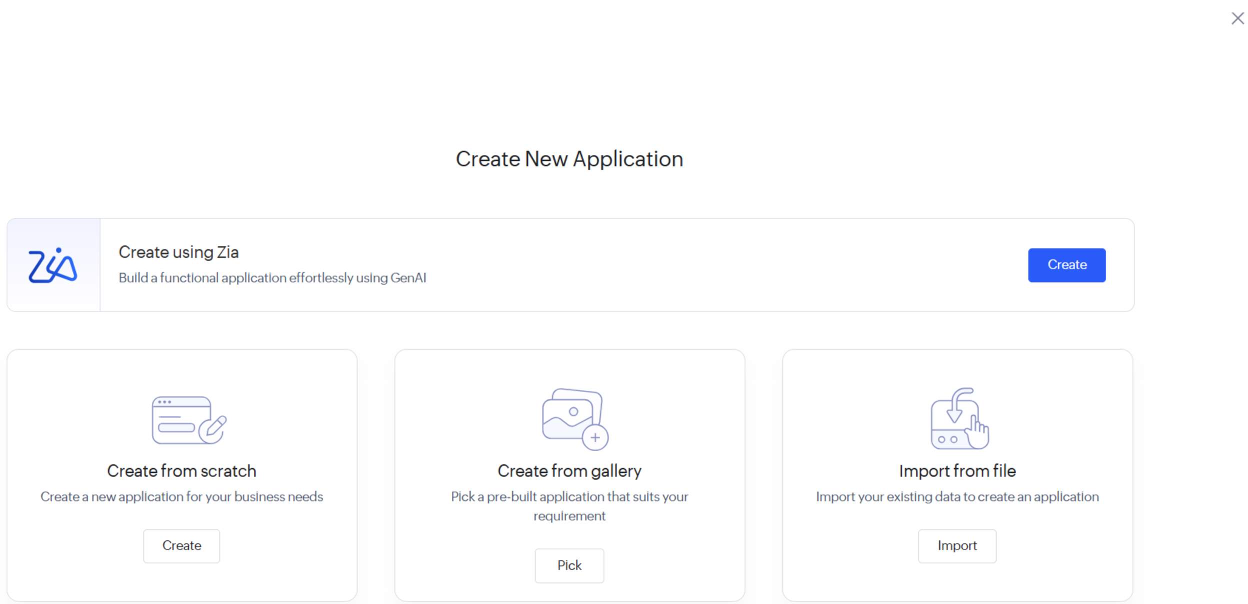
step 7: The software has been created.

ALGORITHM:

STEP 1: GOTO ZOHO.COM

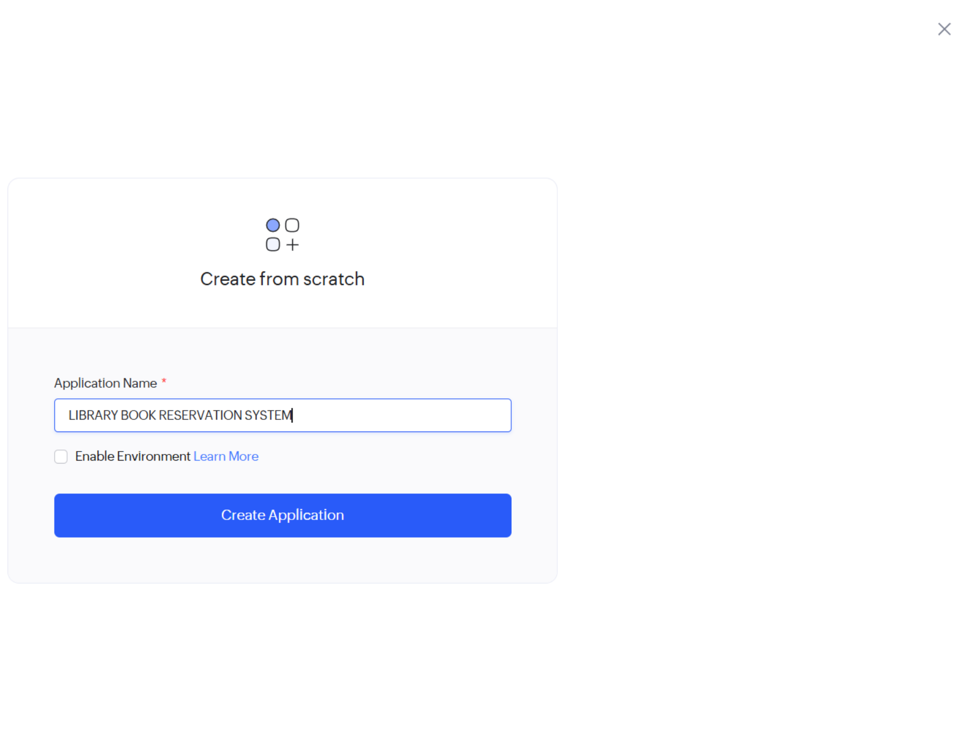


STEP 2: CREATE FROM SCRATCH

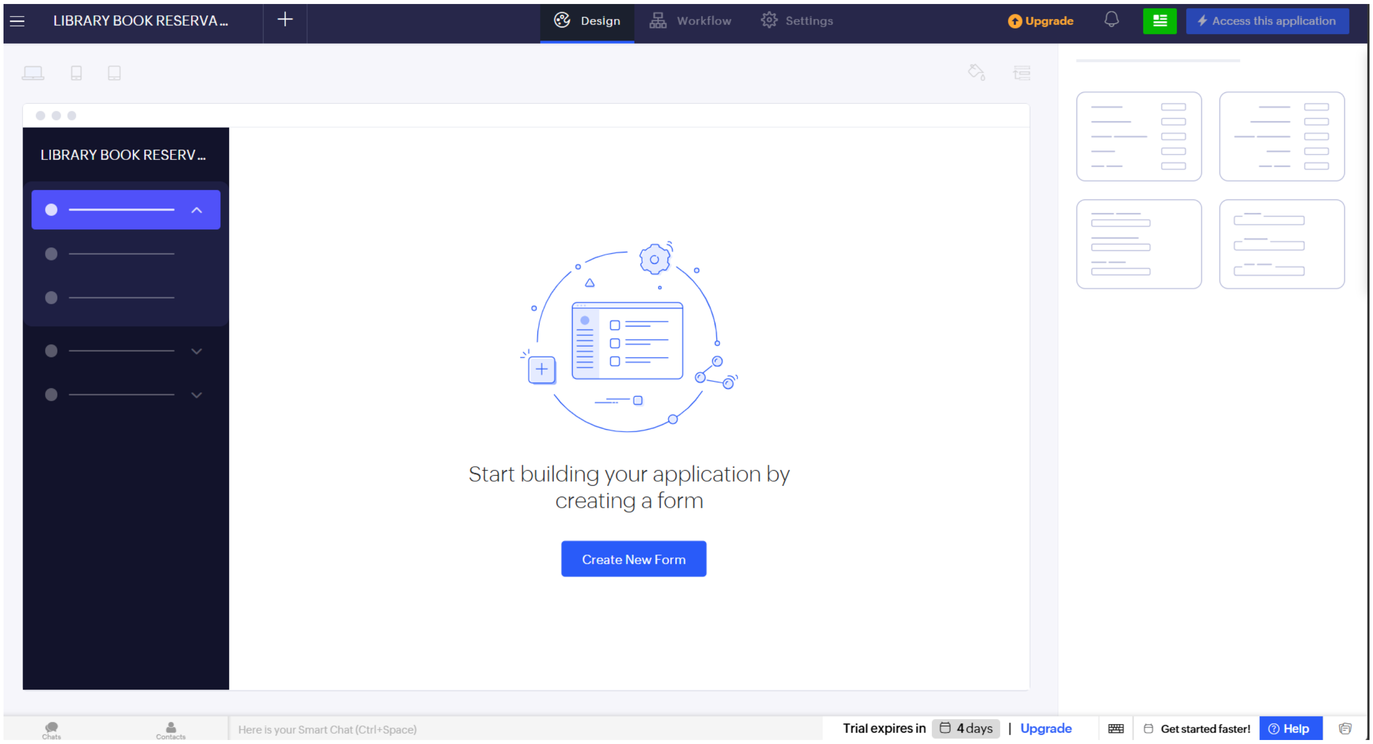


STEP 3: ENTER APPLICATION NAME

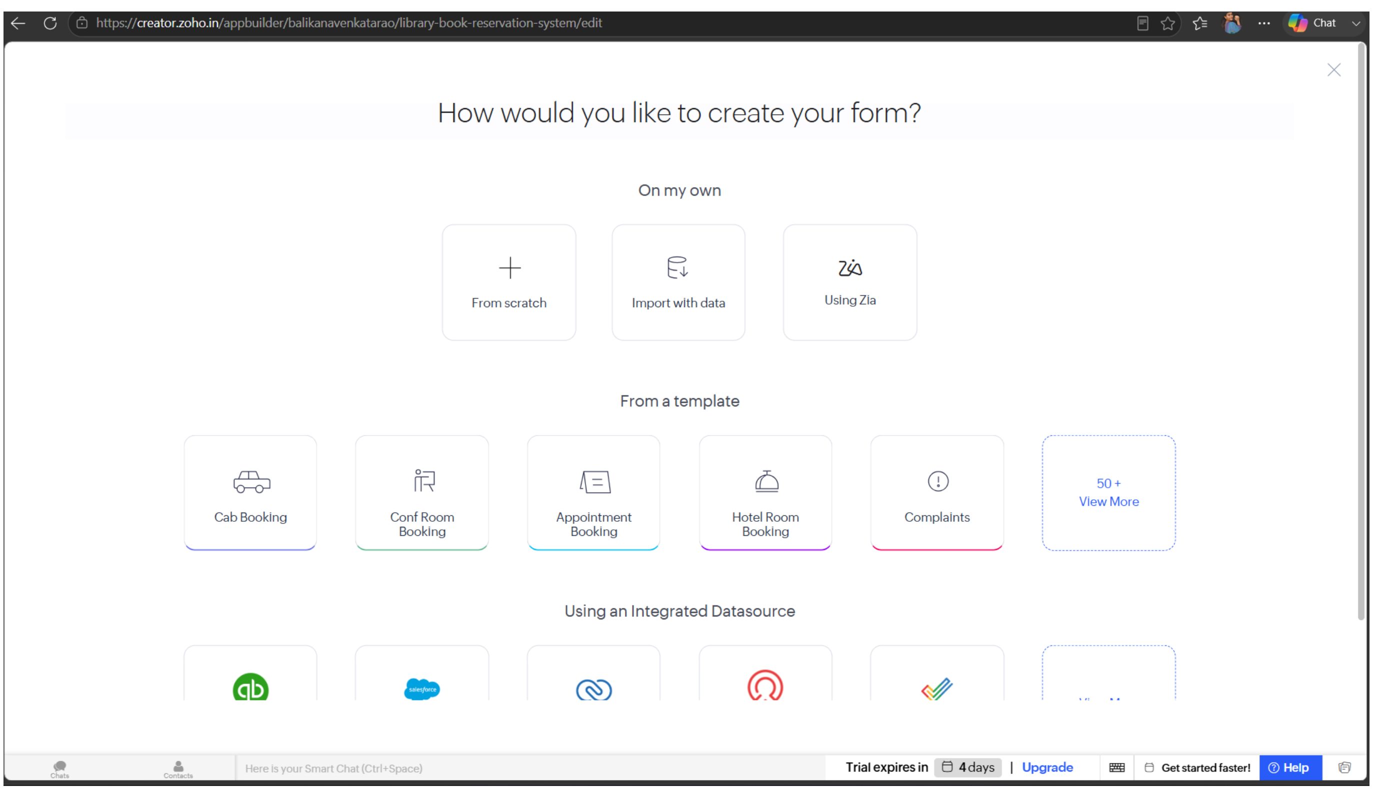
-



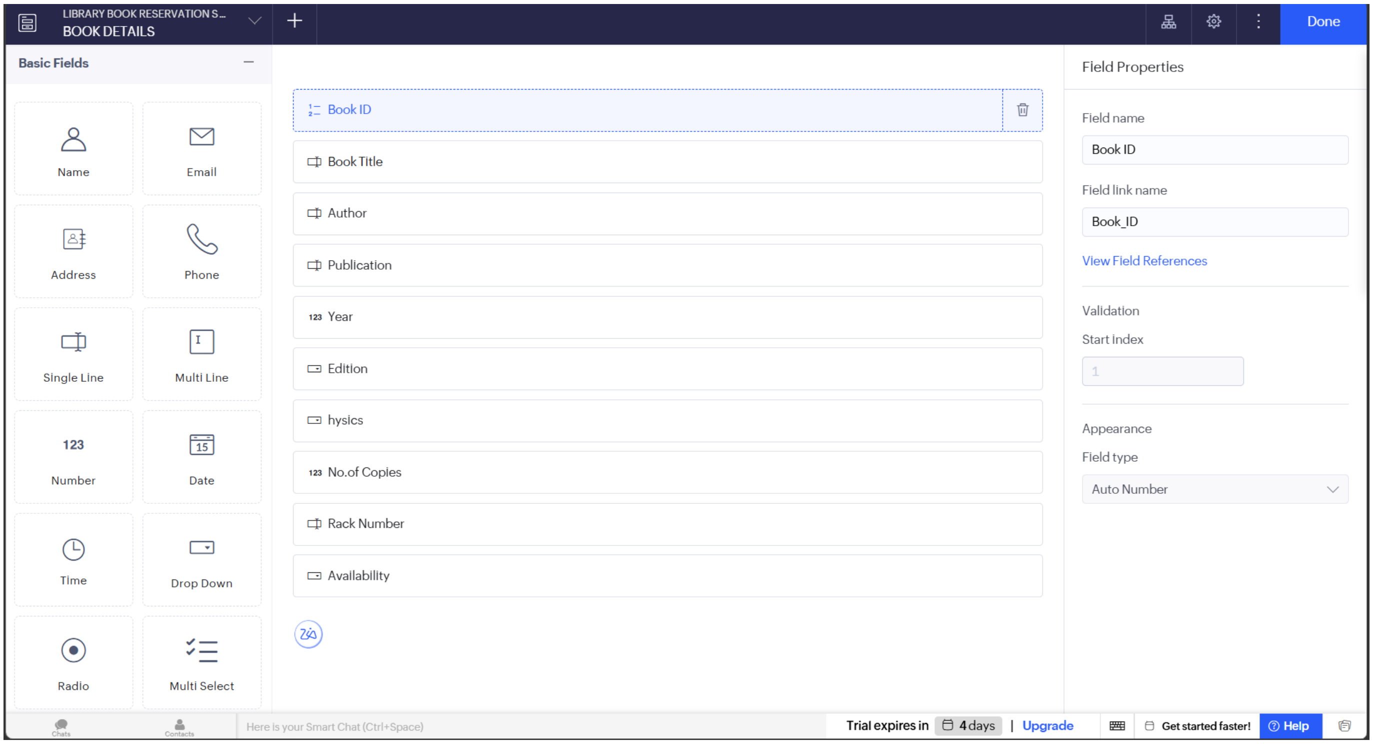
STEP 4: CREATED NEW APPLICATION



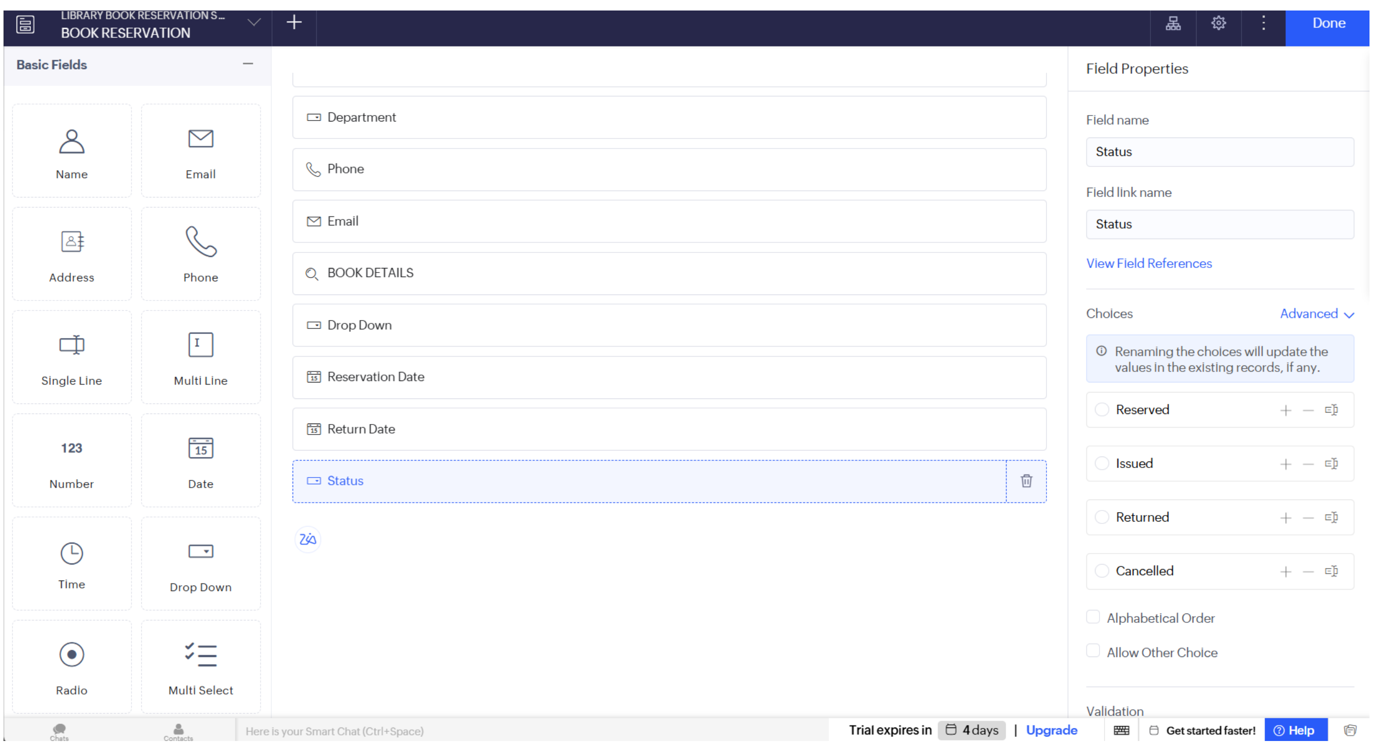
STEP 5: SELECT ONE FORM



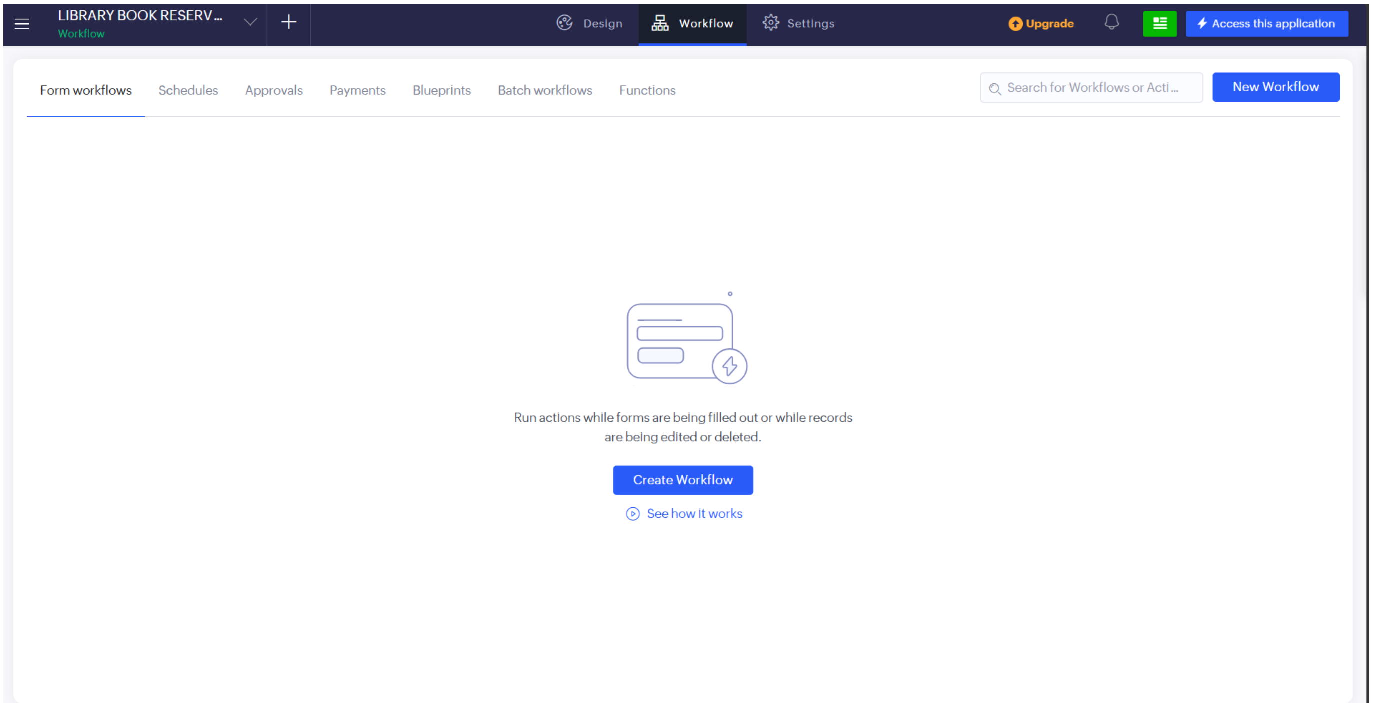
STEP 6: CREATE BOOK DETAILS FORM



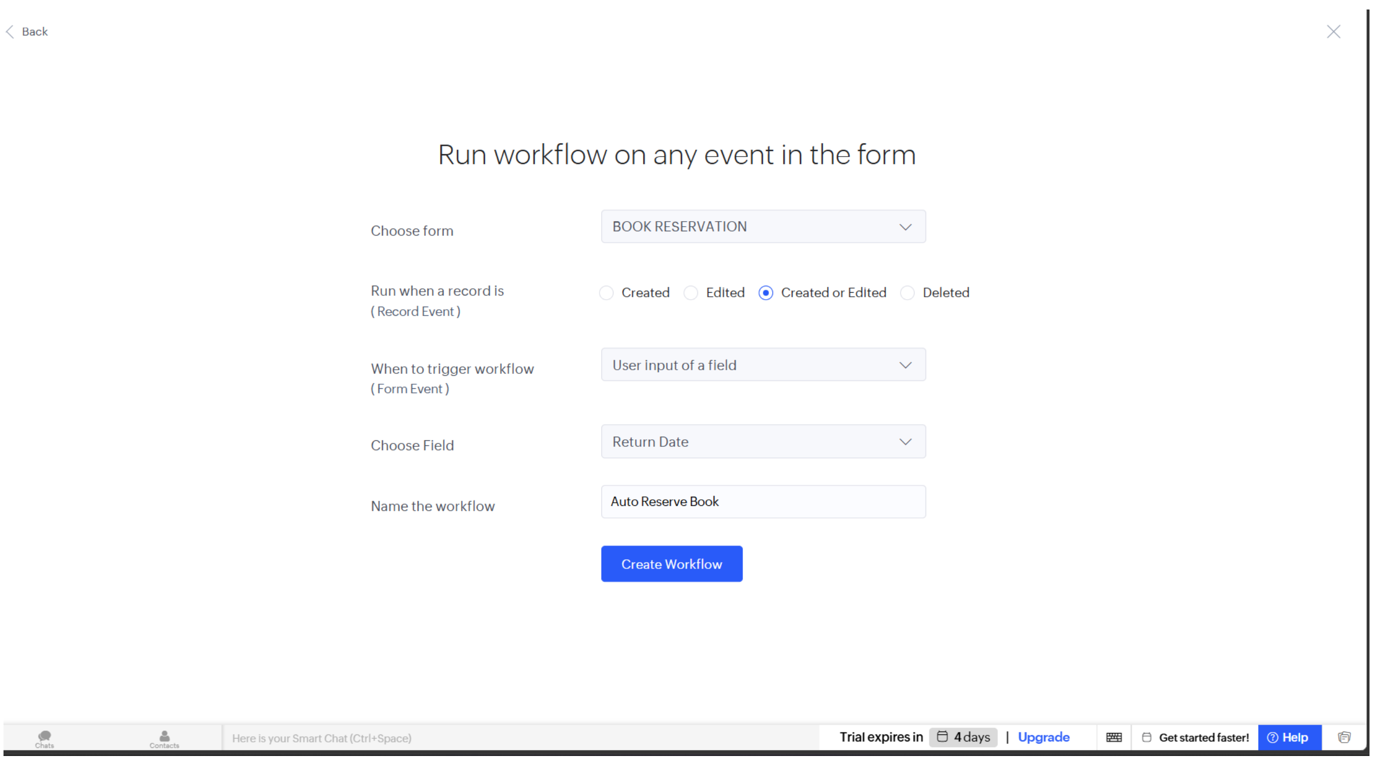
STEP 7: CREATE BOOK RESERVATION FORM



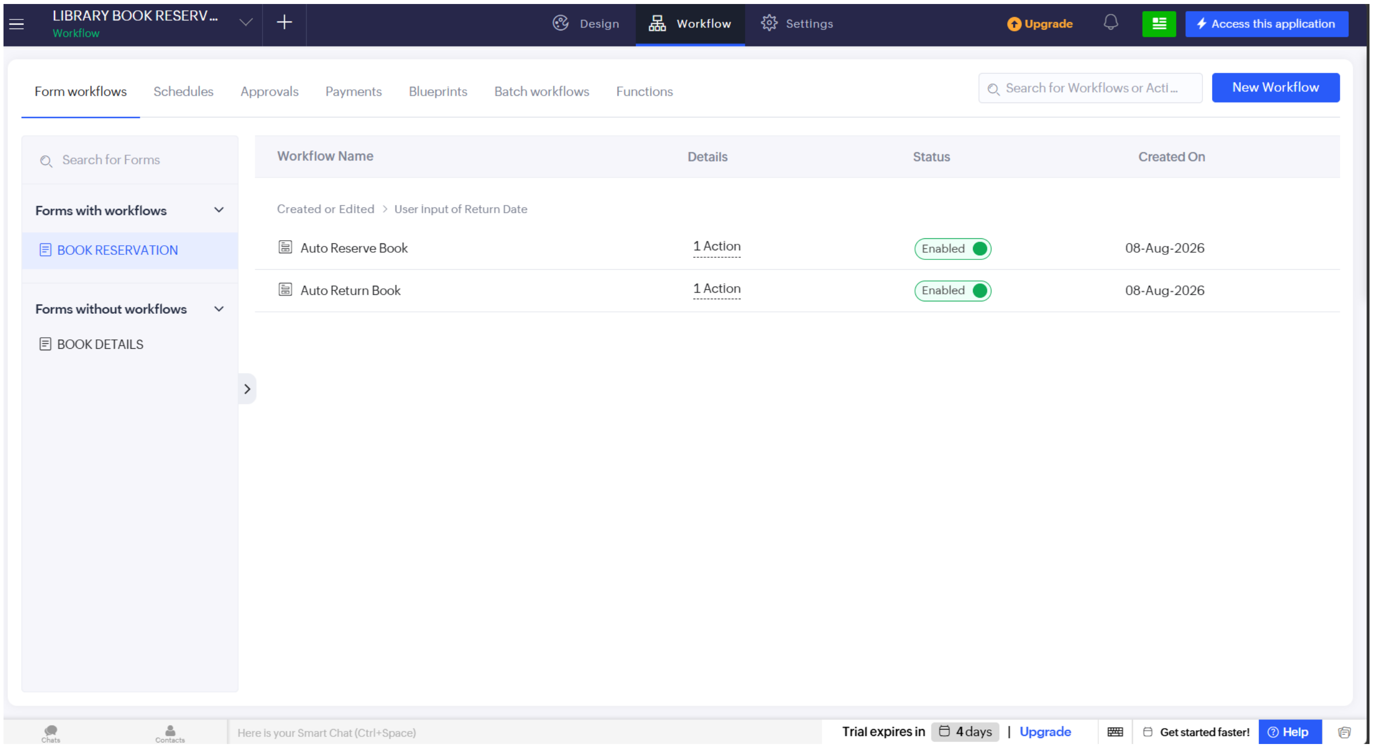
STEP 8: CREATE WORKFLOW



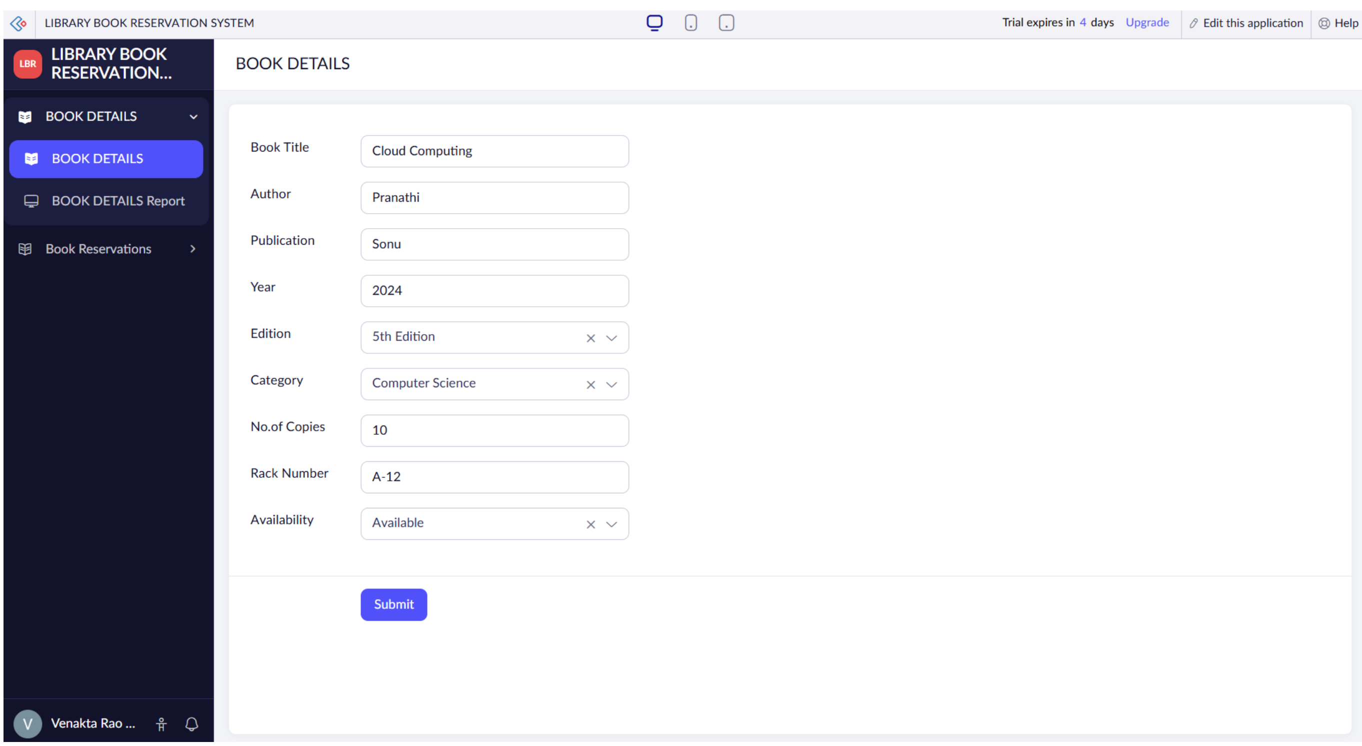
STEP 9: CREATE RUN WORKFLOW ON ANY EVENT IN THE FORM



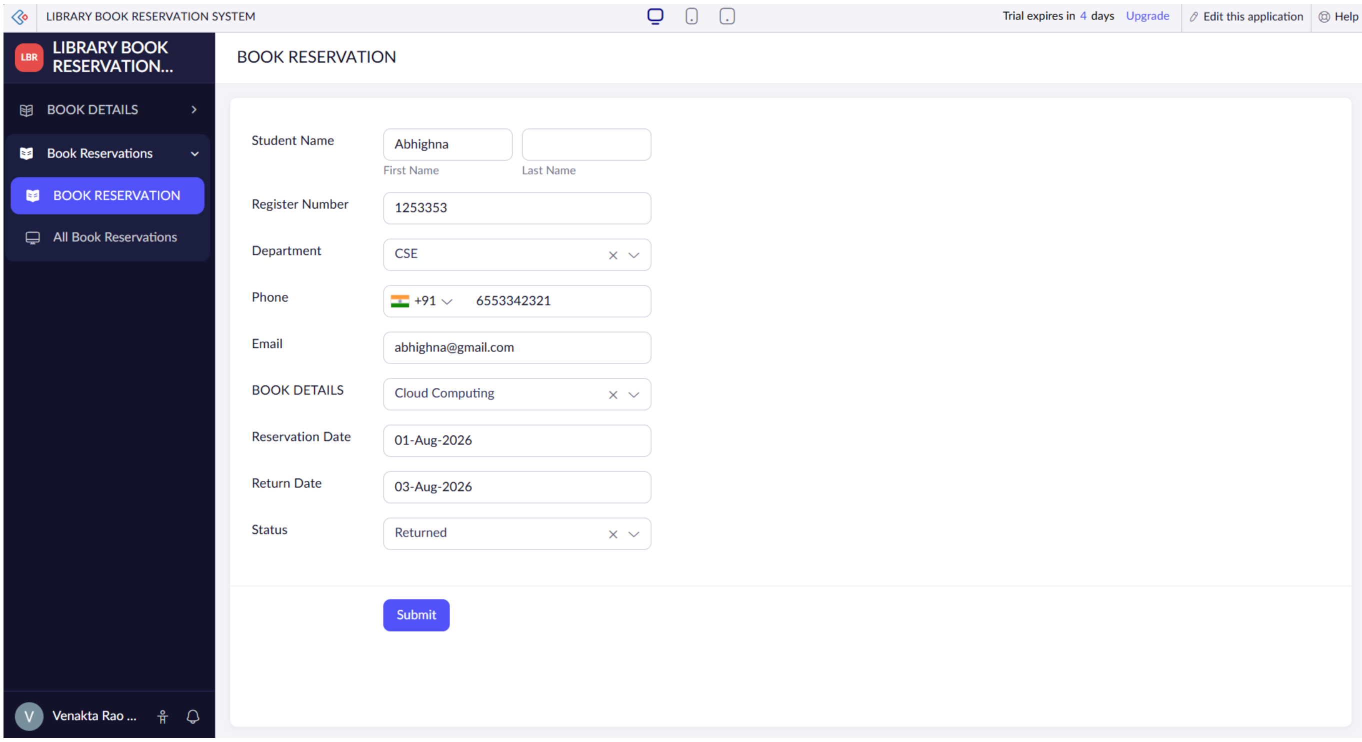
STEP 10: WORKFLOW CREATED FOR BOOK RESERVATION AND ACCESS THE APPLICATION



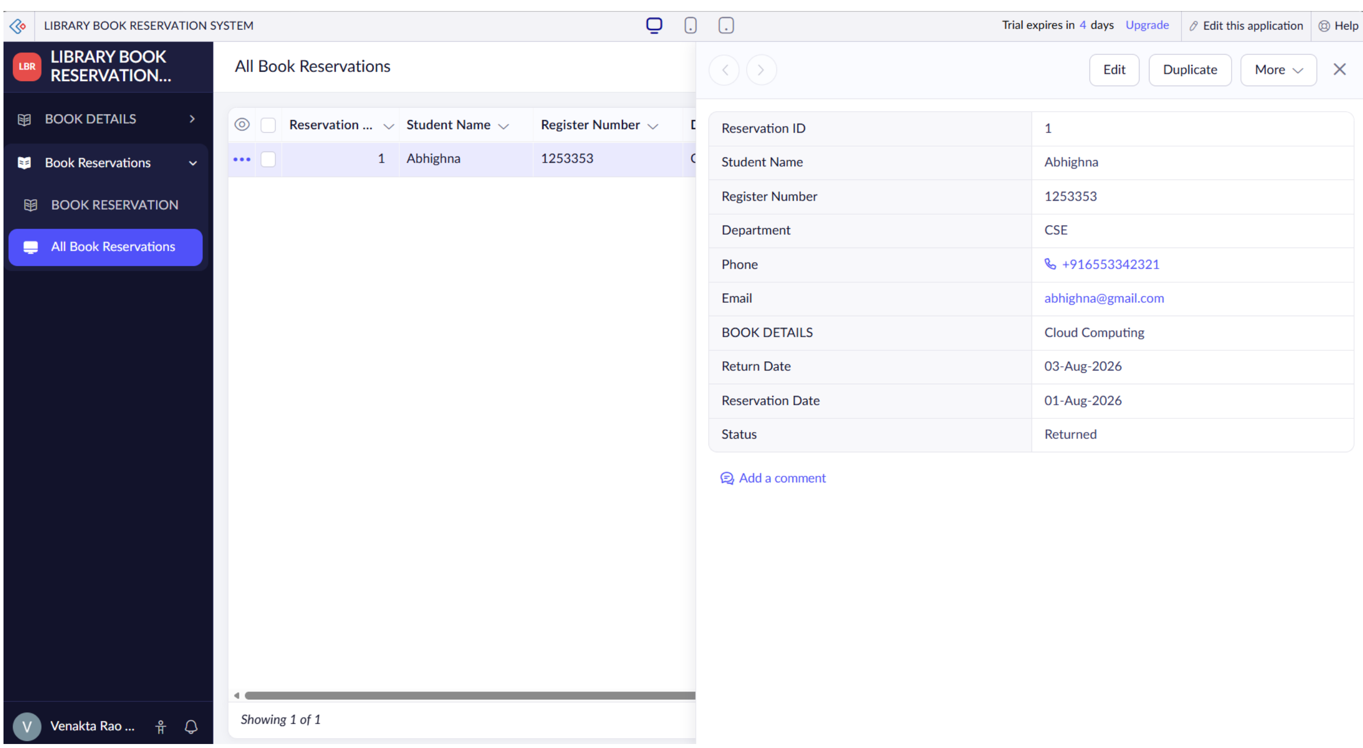
STEP 11: IT OPENS THE BOOK DETAILS FORM AND THEN FILL THE FORM AND CLICK SUBMIT



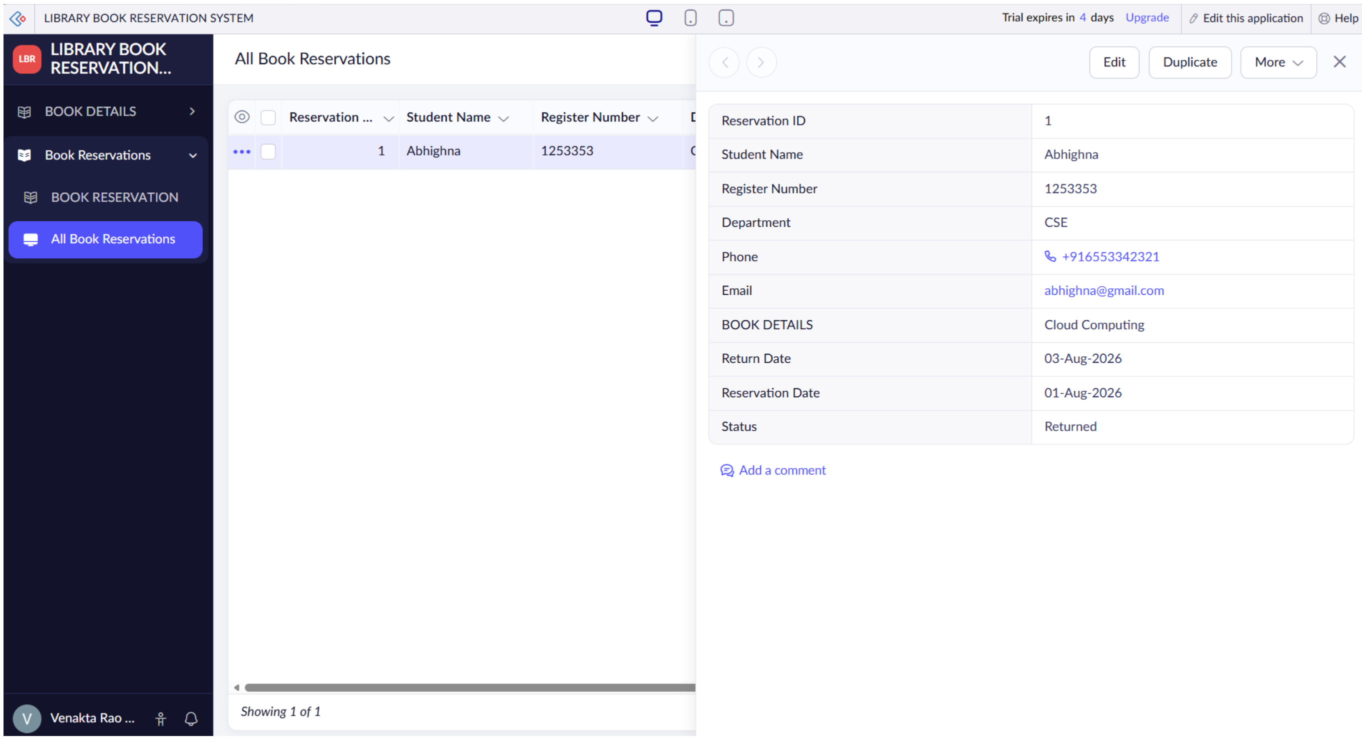
STEP 12: AND FILL THE FORM OF BOOK RESERVATION DETAILS AND THEN CLICK SUBMIT



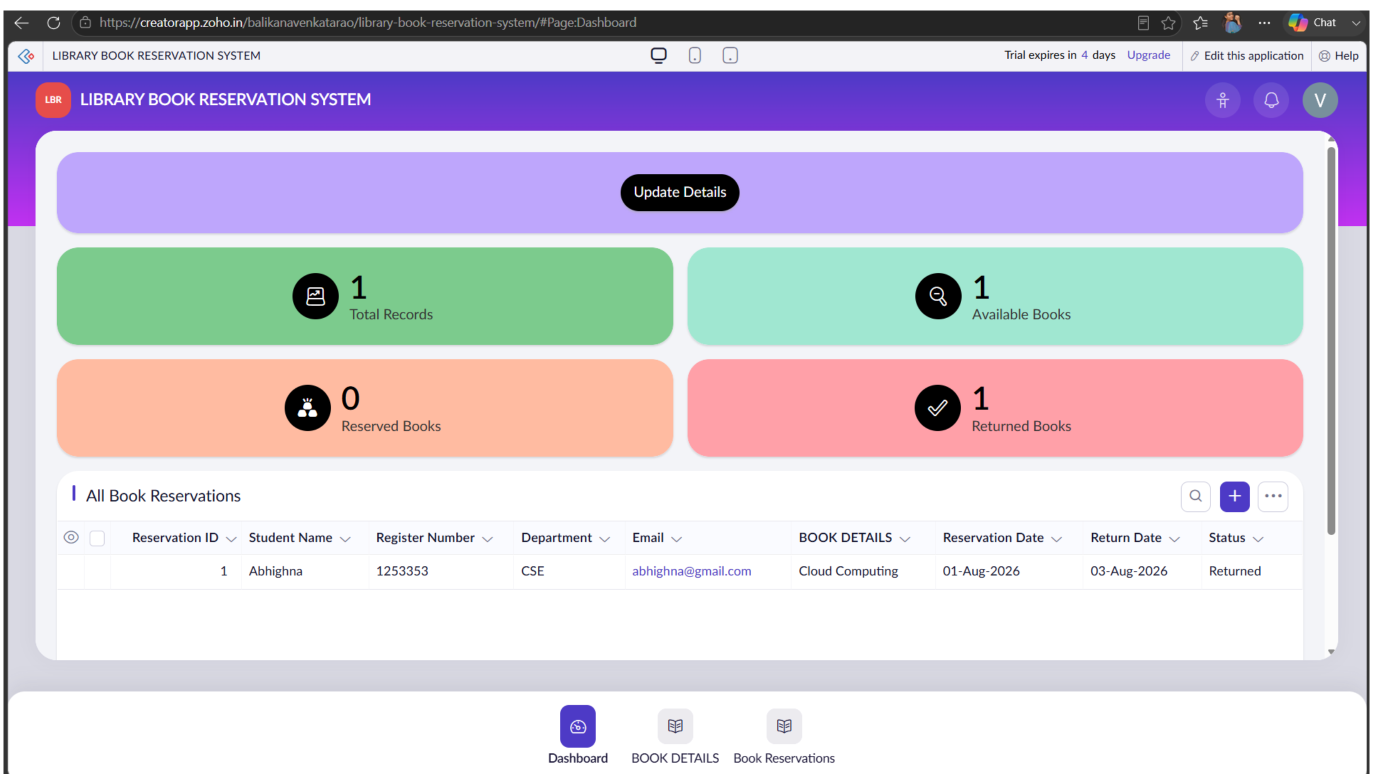
STEP 13: THE DETAILS ARE FINALLY STORED IN BOOK RESERVATION REPORT



STEP 14: FINALLY IT SHOWS THAT BOOK IS RETURNED OR RESERVED



STEP 15: DASHBOARD



RESULT: A simple cloud software application for Library Book Reservation System using any Cloud Service Provider is successfully created.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

AIM:

To Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

step 1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step 4: Enter application name as Payroll Processing System.

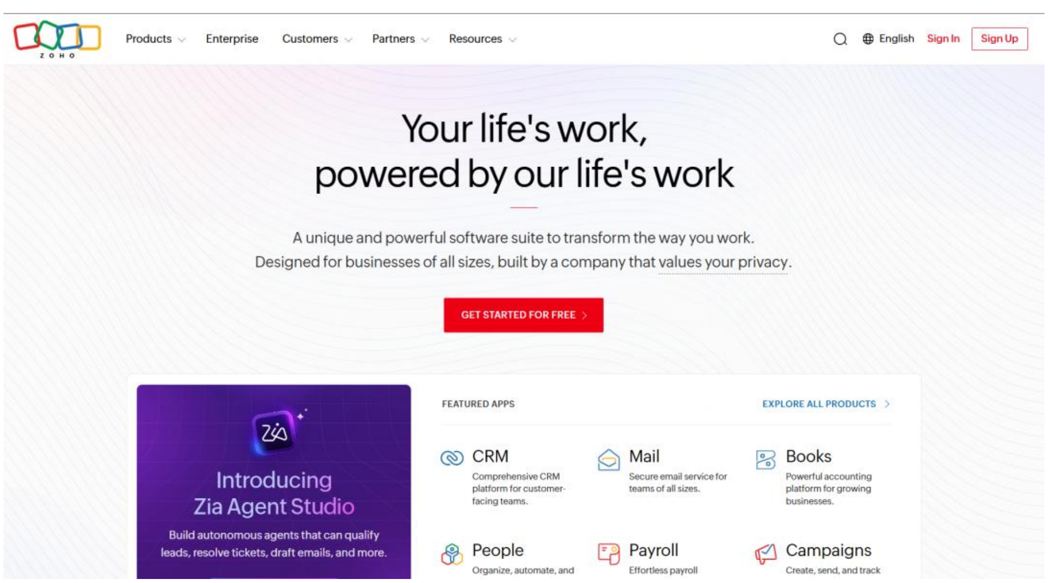
step 5: Created new application as Payroll Processing System.

step 6: Select one form

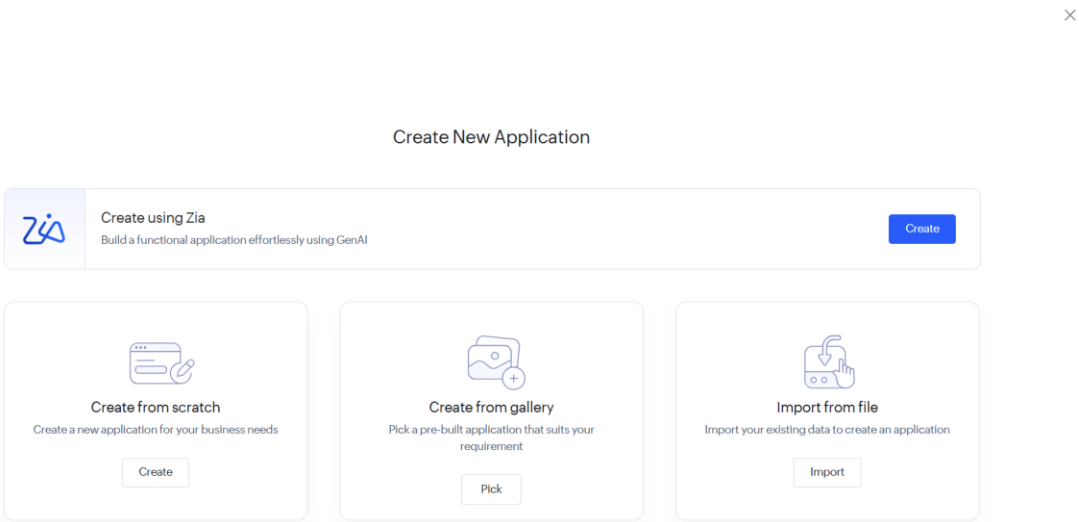
step 7: The software has been created.

ALGORITHM:

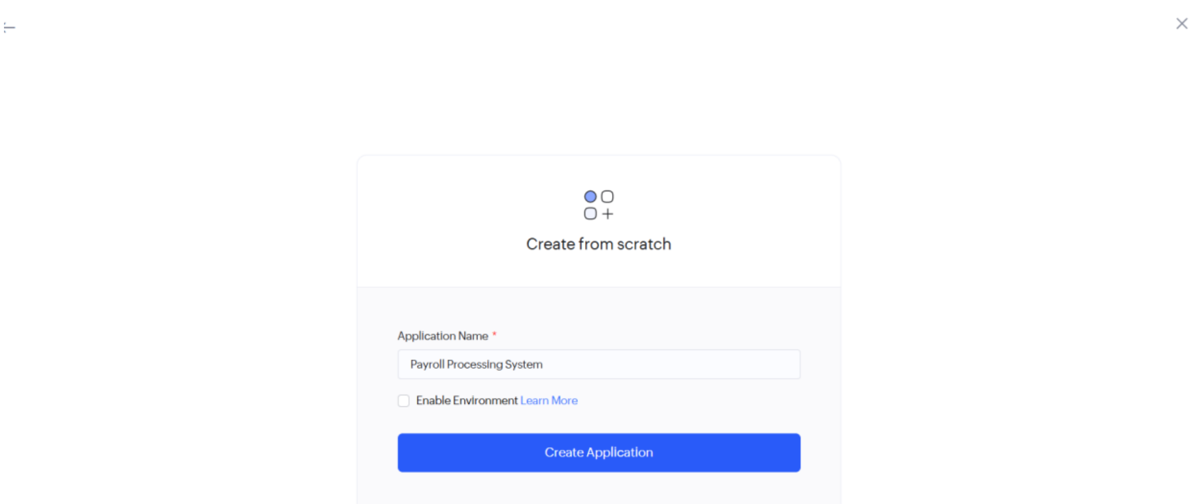
STEP 1: GOTO ZOHO.COM



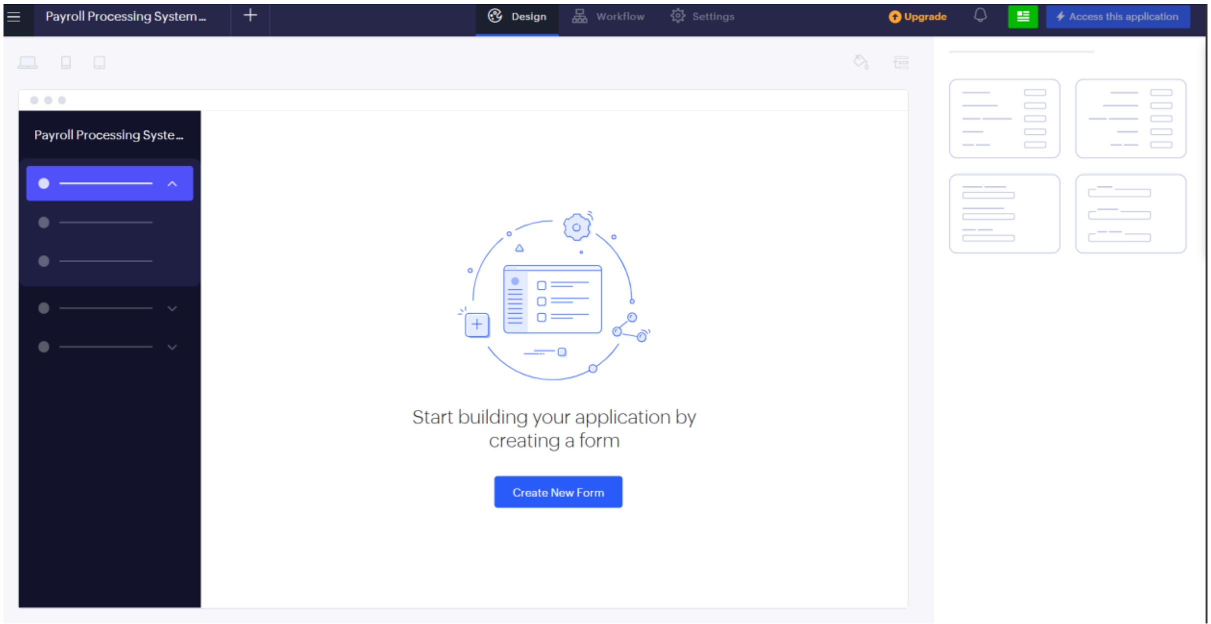
STEP 2: CREATE FROM SCRATCH



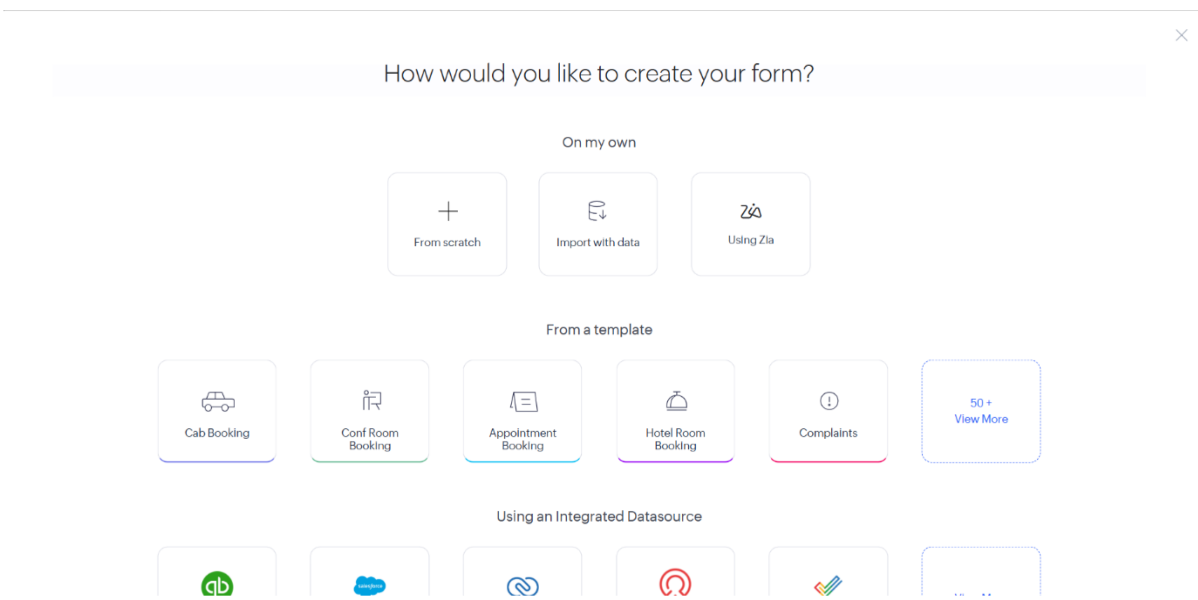
STEP 3: ENTER APPLICATION NAME



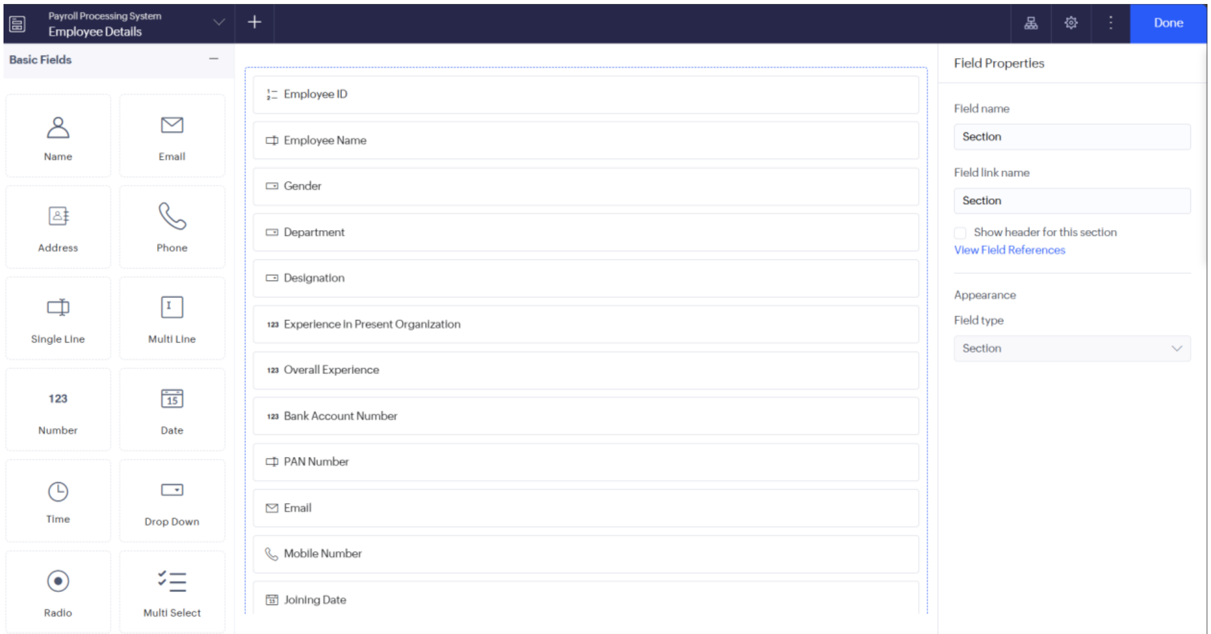
STEP 4: CREATED NEW APPLICATION



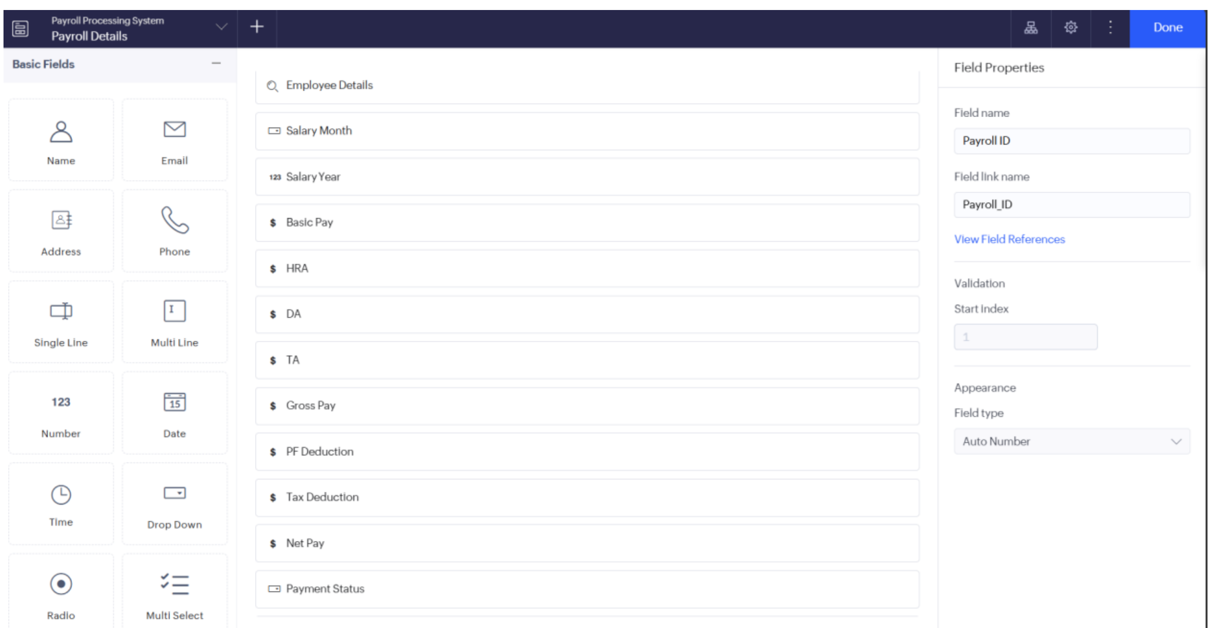
STEP 5: SELECT ONE FORM



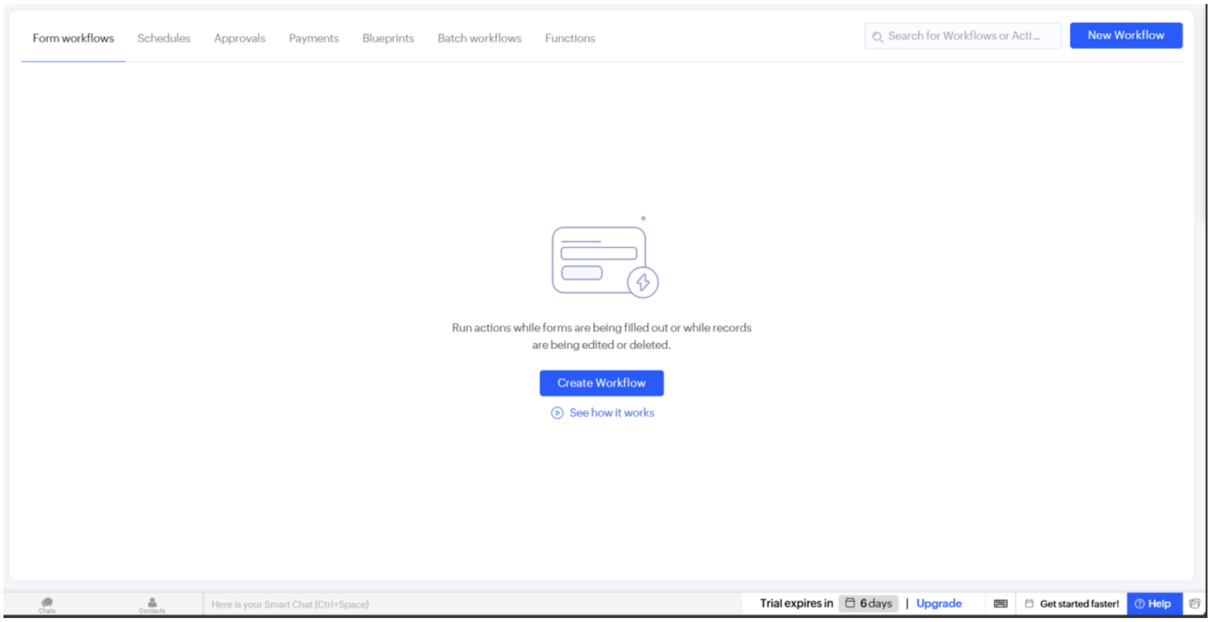
STEP 6: CREATE EMPLOYEE DETAILS FORM



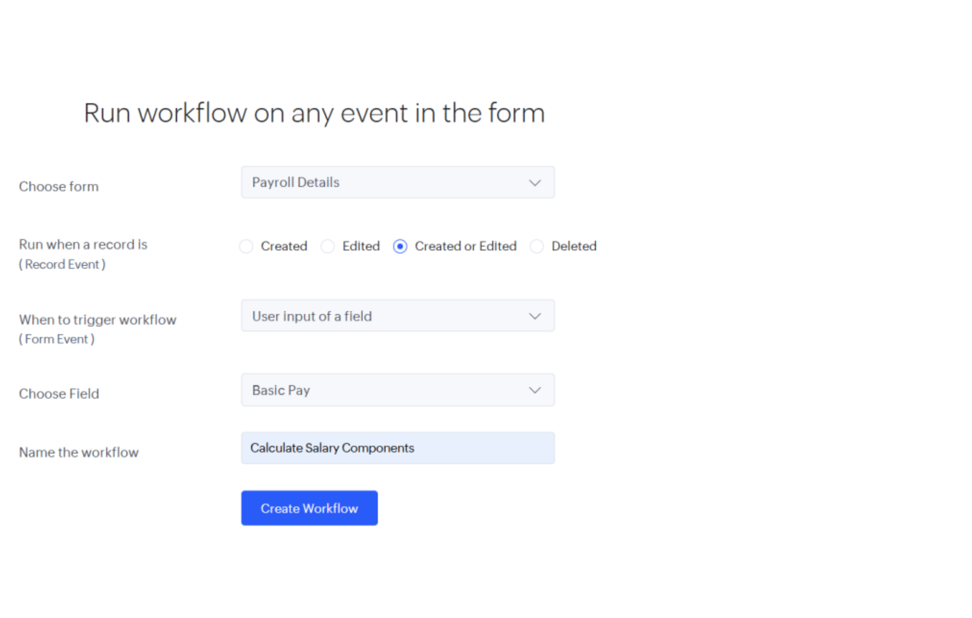
STEP 7: CREATE PAYROLL DETAILS FORM



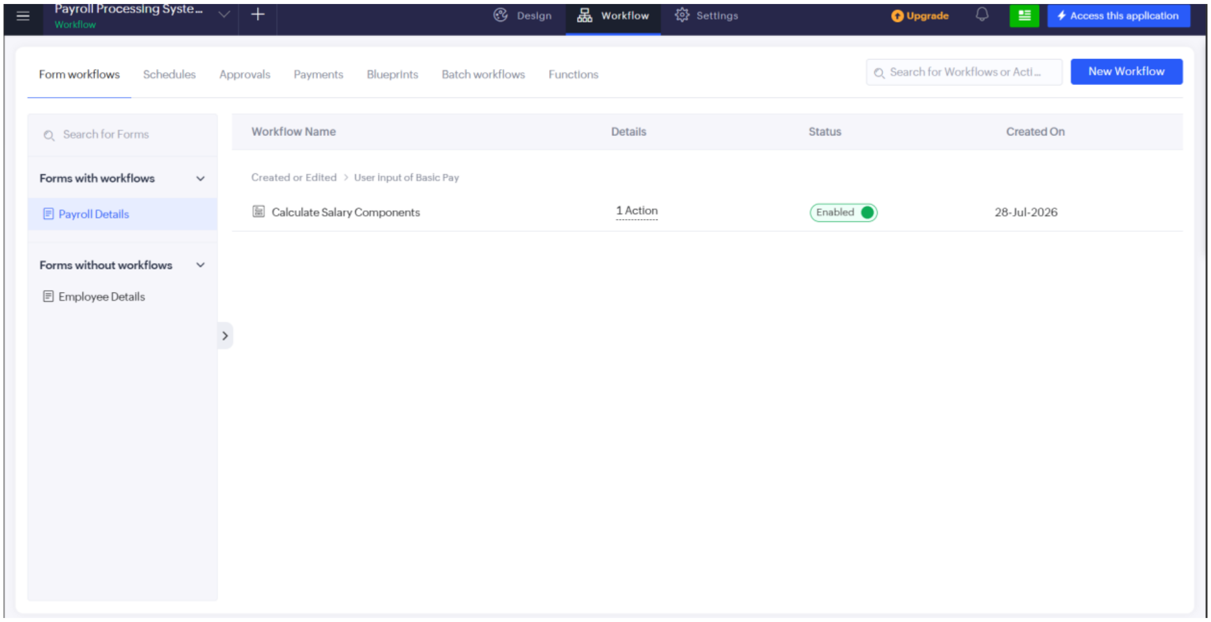
STEP 8: CREATE WORKFLOW



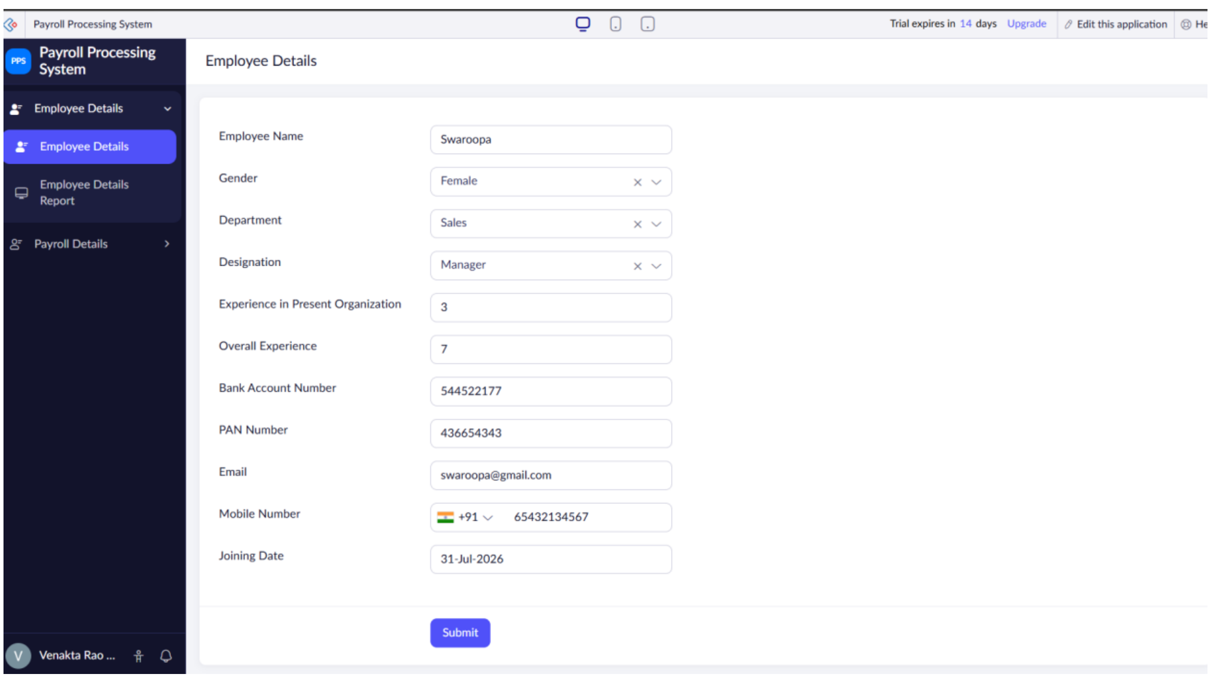
STEP 9: CREATE RUN WORKFLOW ON ANY EVENT IN THE FORM



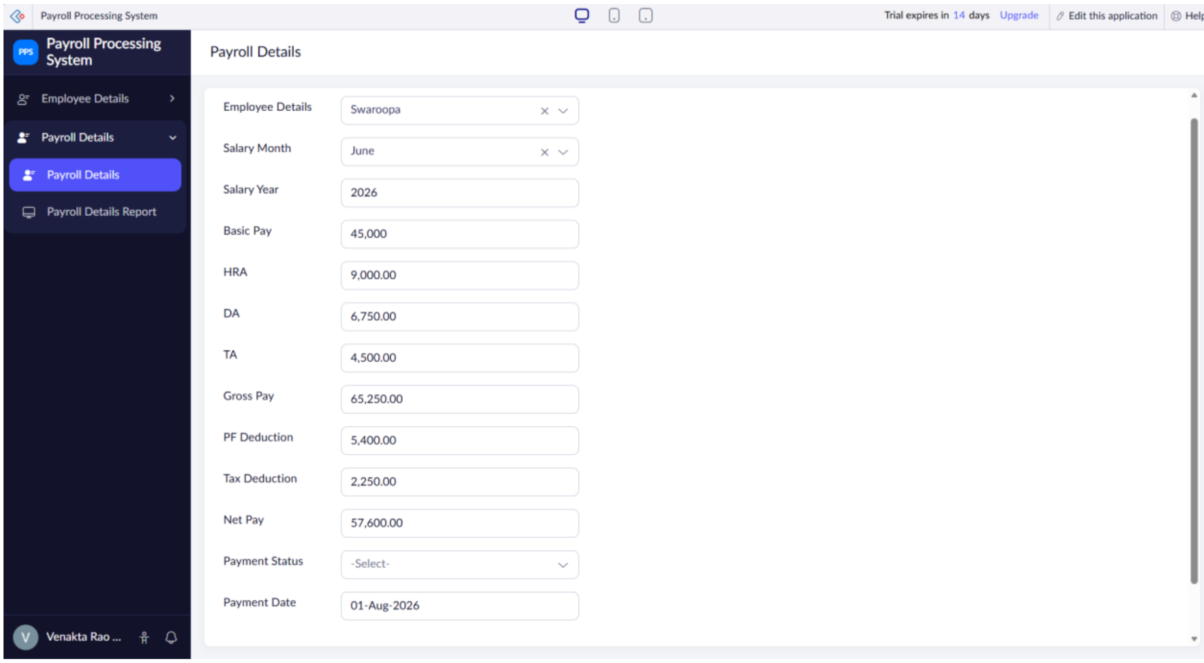
STEP 10: WORKFLOWS CREATED FOR PAYROLL PROCESSING SYSTEM AND ACCESS THE APPLICATION



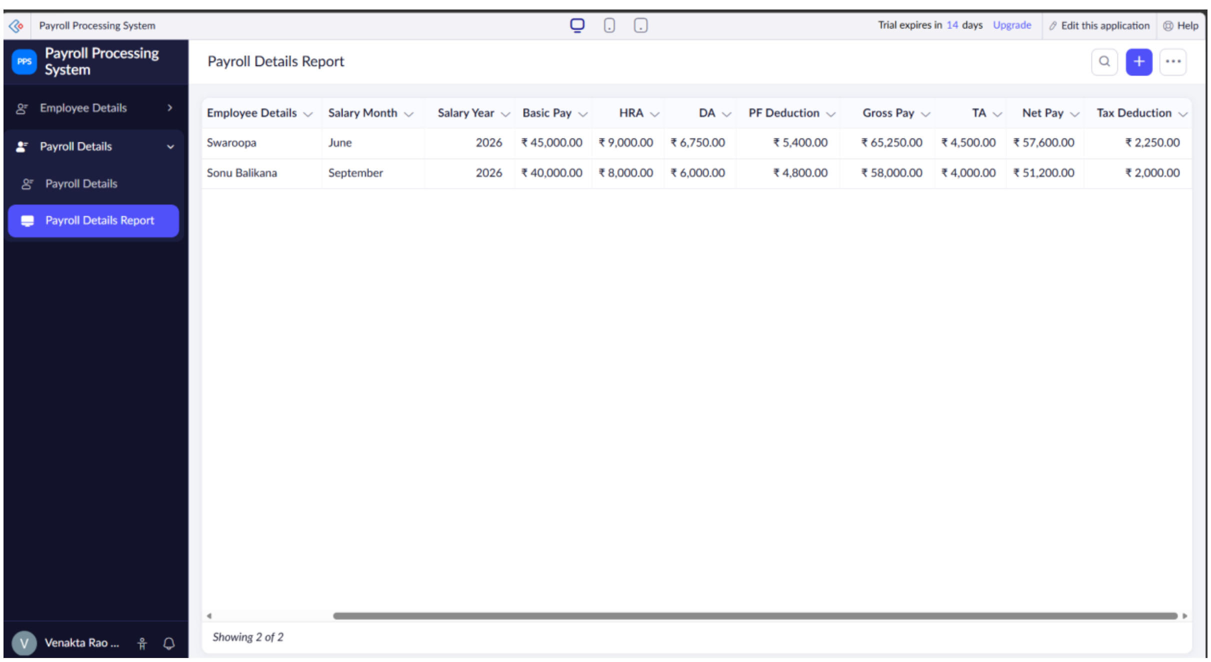
STEP 11: IT OPENS THE EMPLOYEE DETAILS FORM AND THEN FILL THE FORM AND CLICK SUBMIT



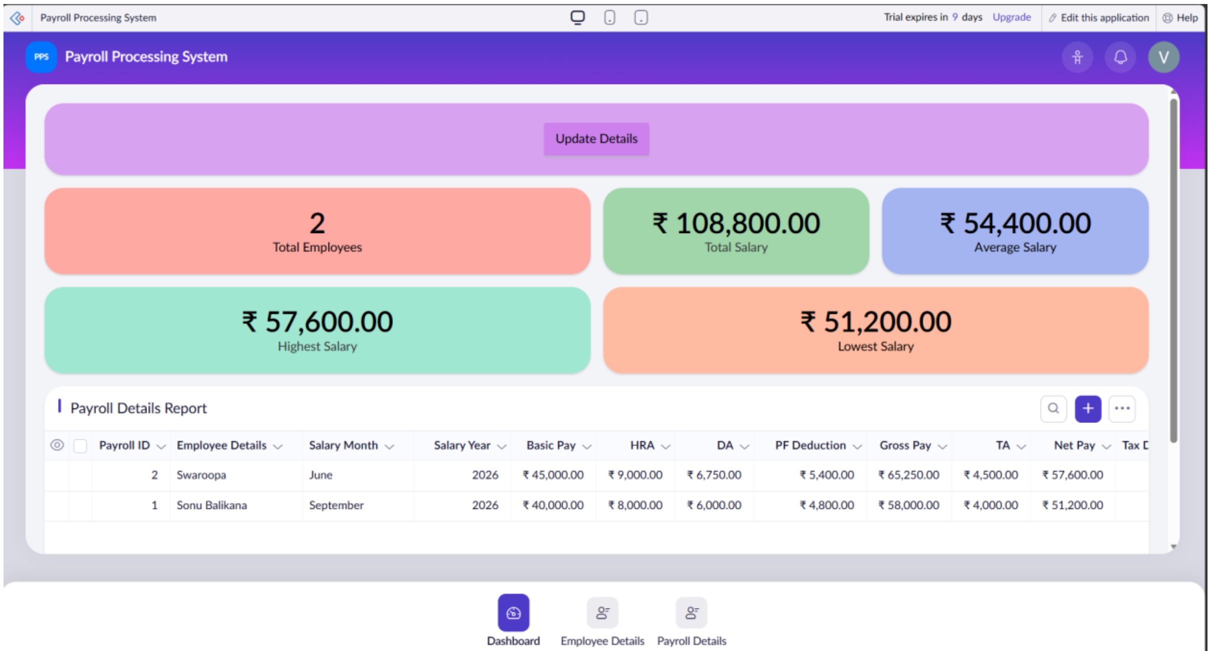
STEP 12: AND FILL THE FORM OF PAYROLL DETAILS AND THEN CLICK SUBMIT



STEP 13: IT AUTOMATICALLY CALCULATES THE HRA,DA,PFA DEDUCTION,GROSS PAY,TA,NET PAY,TAX DEDUCTION



STEP 14: DASHBOARD



RESULT: A simple cloud software application for Payroll Processing System using any Cloud Service Provider is successfully created.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux) and creating a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software

PROCEDURE:

STEP 1:Dowload VMware workstation and installed as type 2hypervisor.

STEP2:Dowload ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

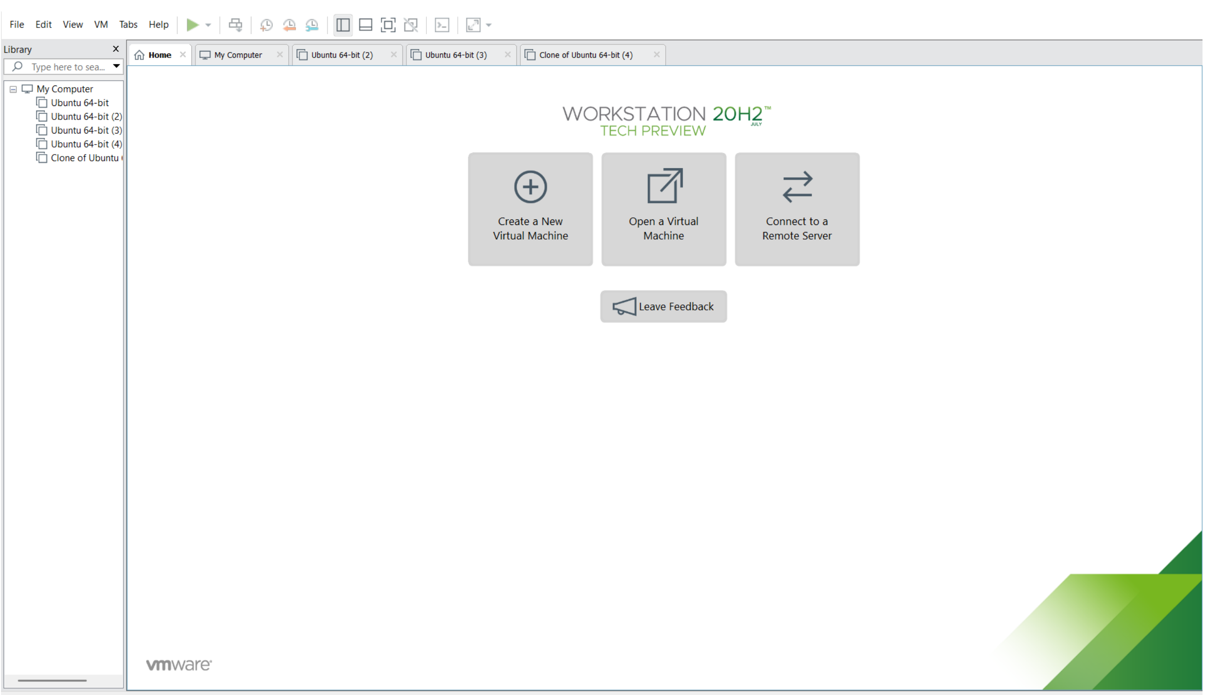
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

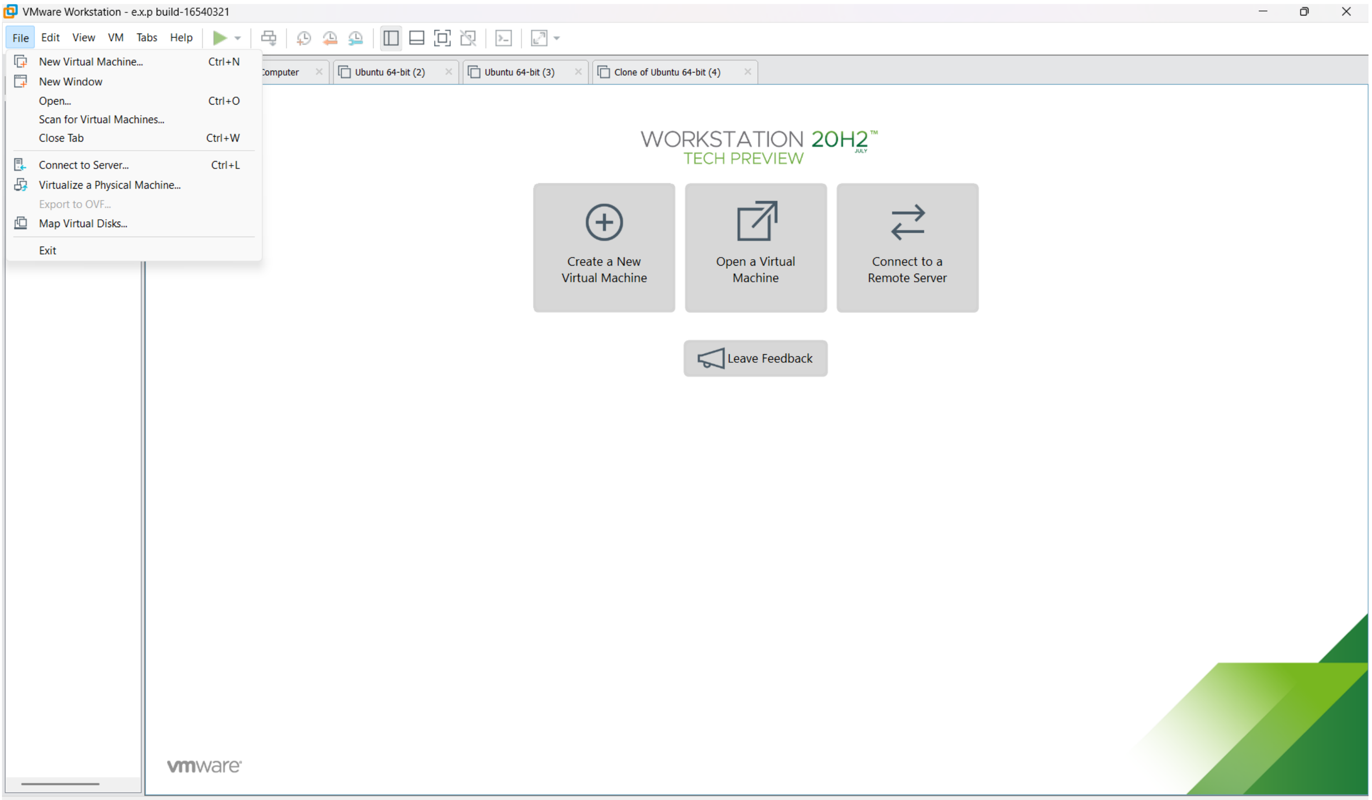
STEP 6: Launch the VM.

ALGORITHM:

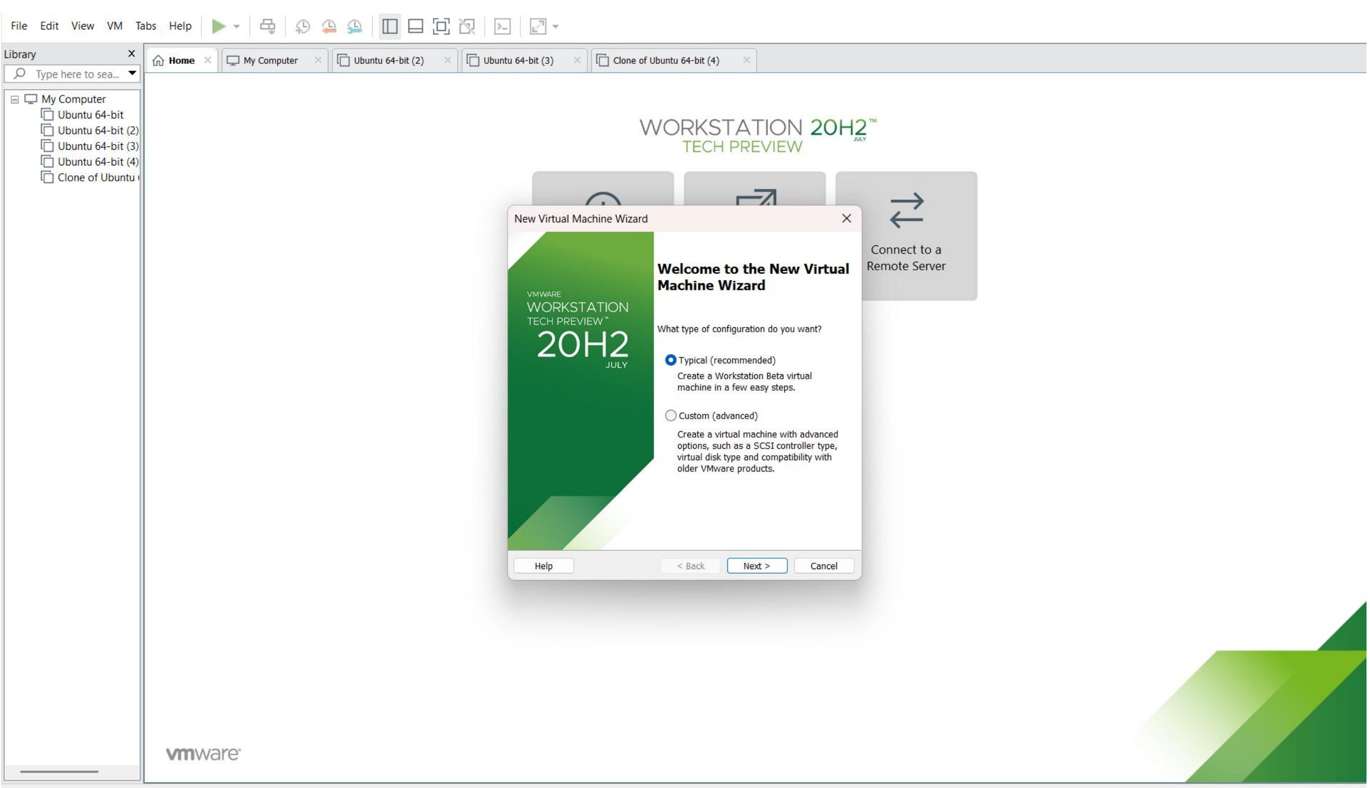
STEP 1: DOWLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2HYPERVISOR

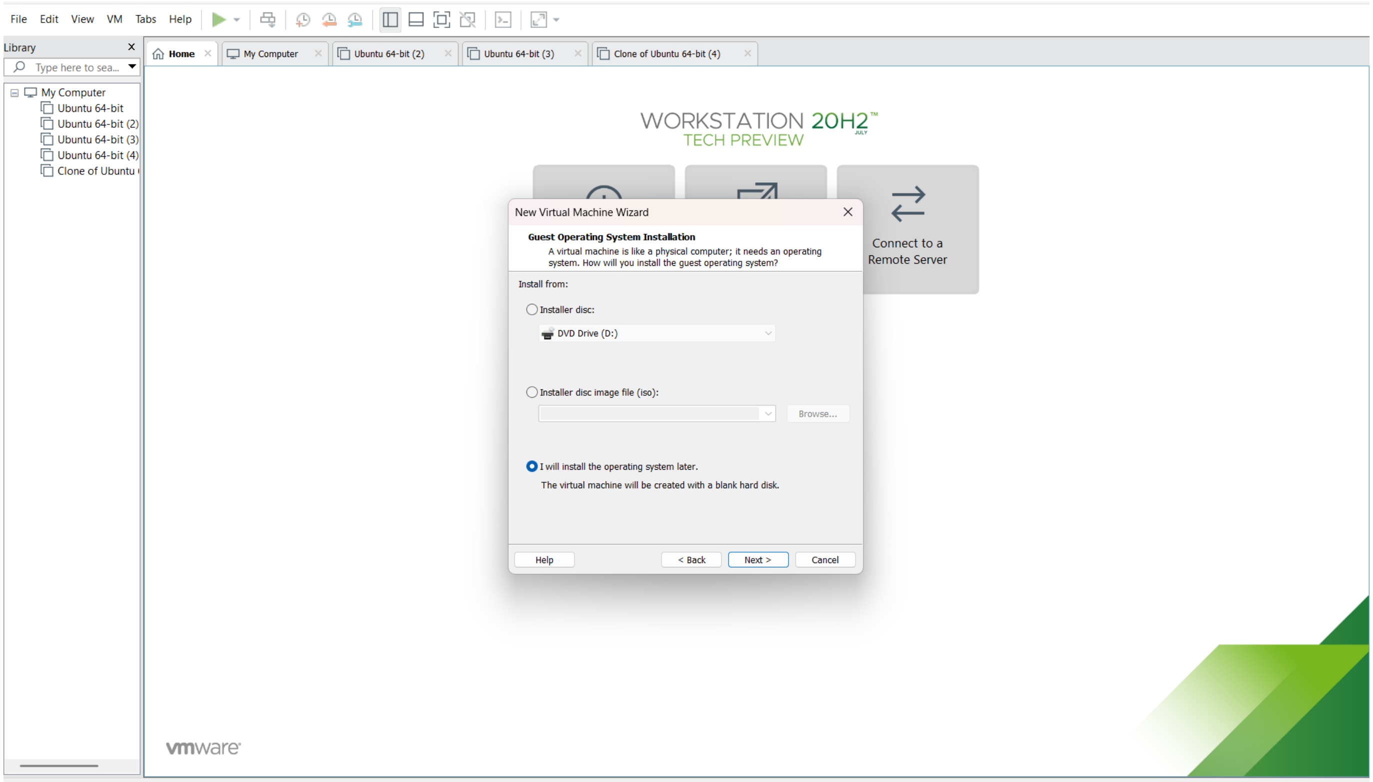


STEP 2: IN VMWARE WORKSTATION->CREATE NEW VM

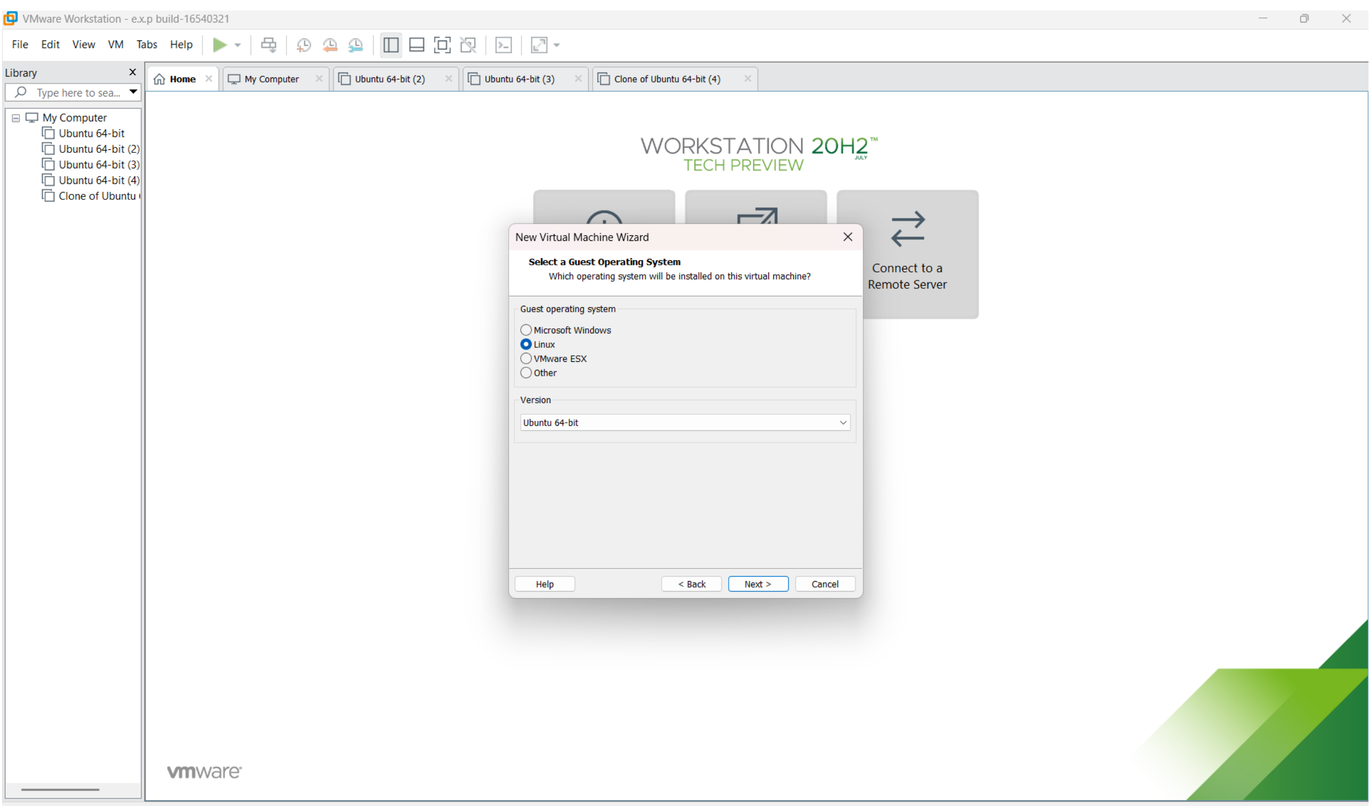


STEP 3: DO THE BASIC CONFIGURATION SETTINGS.

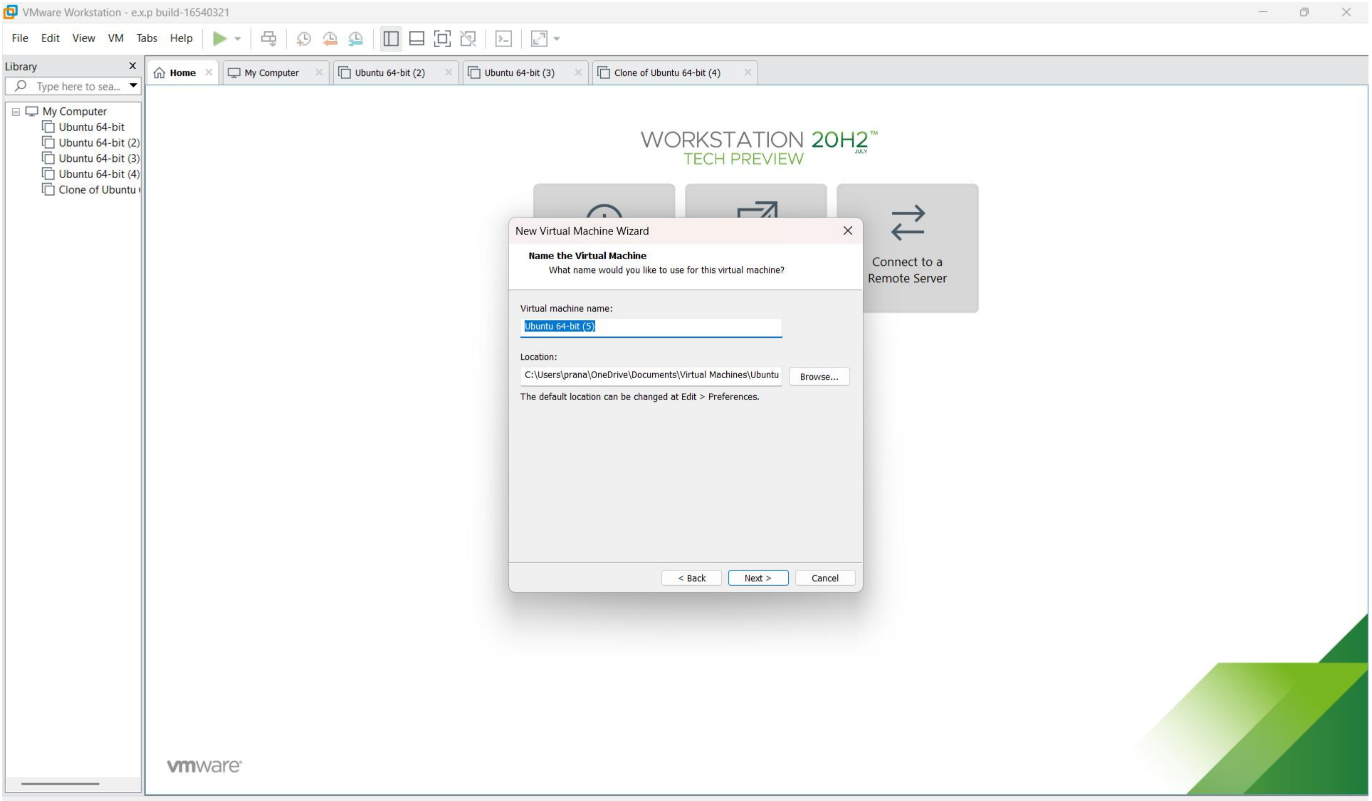
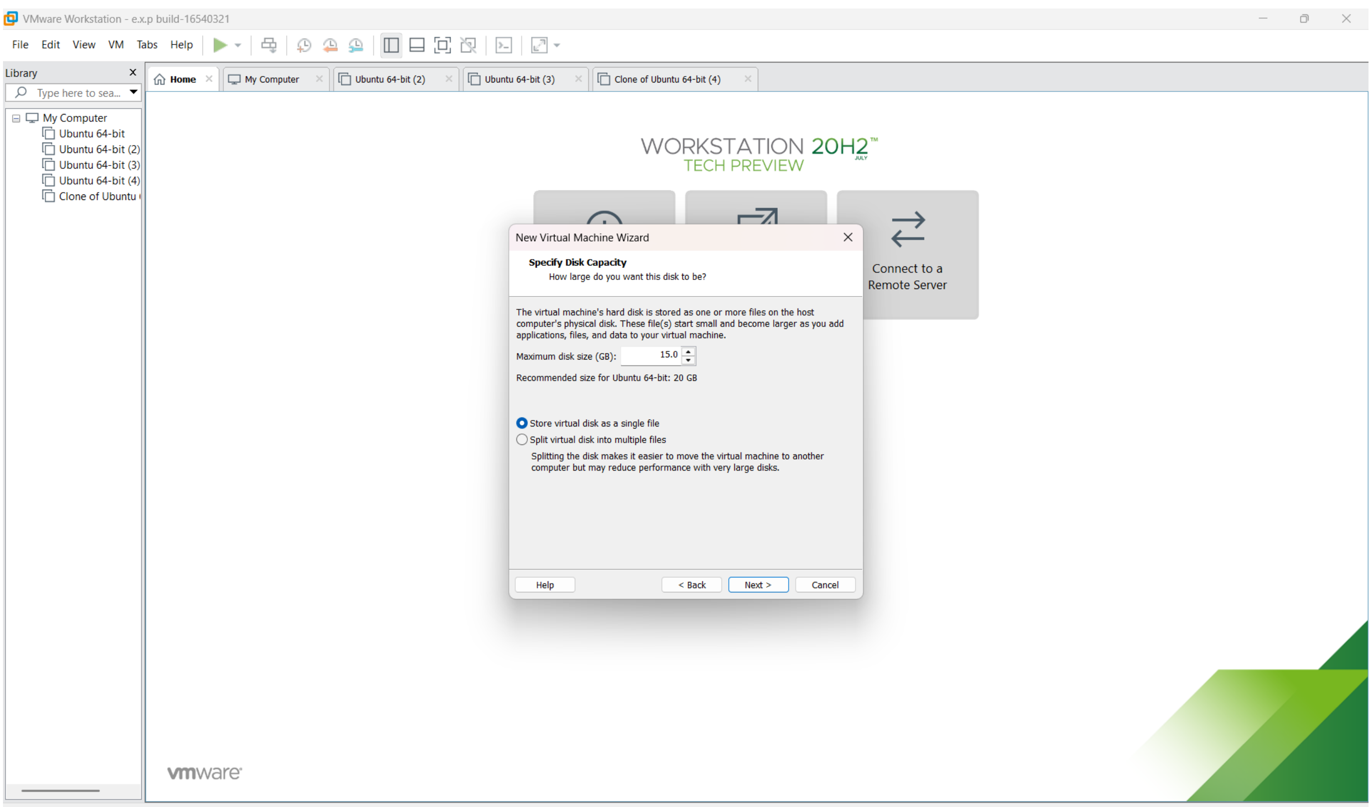


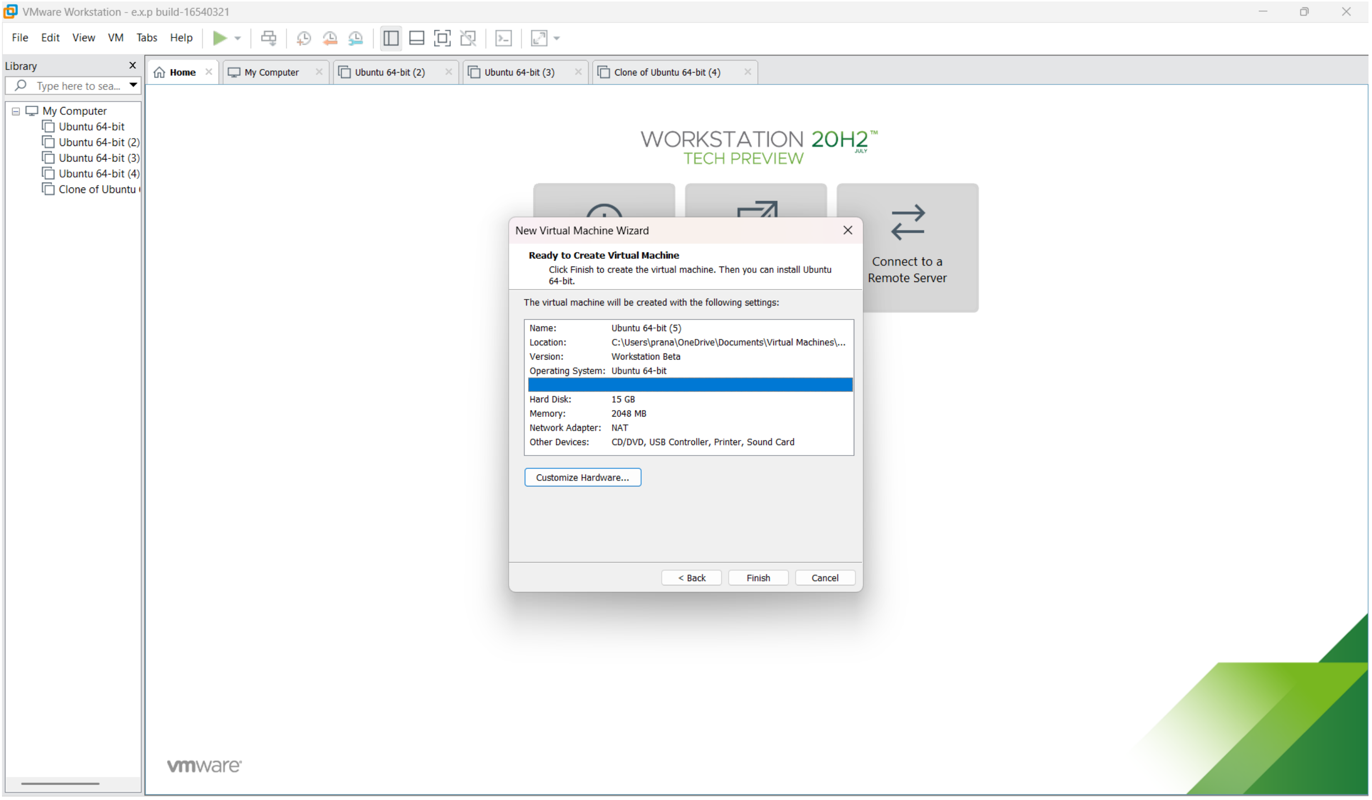


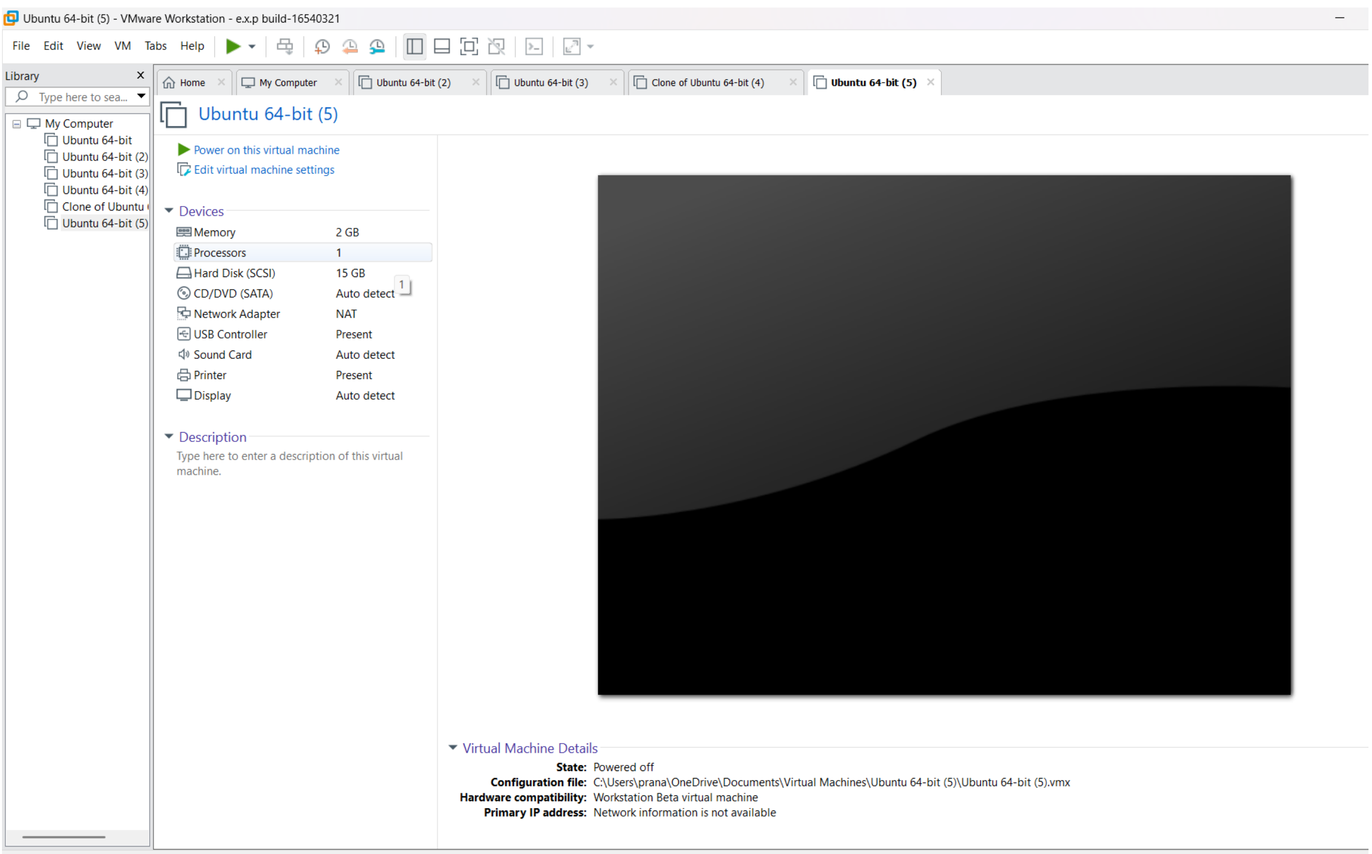
New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
DVD Drive (D:)
Installer disc image file (iso):
Browse...
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel

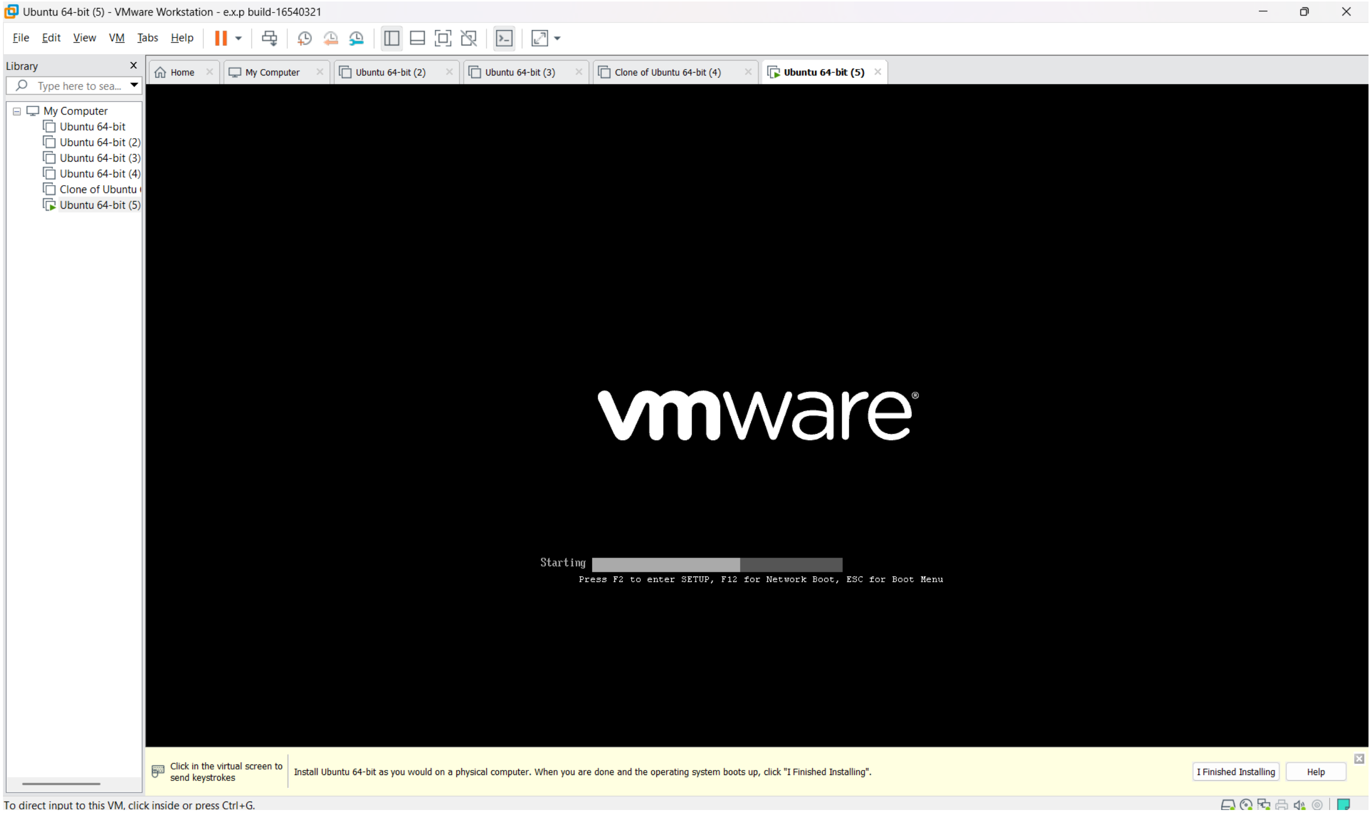


New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Ubuntu 64-bit
Help
< Back
Next >
Cancel


VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Ubuntu 64-bit (3)
Ubuntu 64-bit (4)
Clone of Ubuntu
Home
My Computer
Ubuntu 64-bit (2)
Ubuntu 64-bit (3)
Clone of Ubuntu 64-bit (4)
WORKSTATION 20H2™
TECH PREVIEW
Connect to a Remote Server
New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Ubuntu 64-bit (5)
Location:
C:\Users\prana\OneDrive\Documents\Virtual Machines\Ubuntu
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel
vmware

VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Ubuntu 64-bit (3)
Ubuntu 64-bit (4)
Clone of Ubuntu
Home
My Computer
Ubuntu 64-bit (2)
Ubuntu 64-bit (3)
Clone of Ubuntu 64-bit (4)
WORKSTATION 20H2™
TECH PREVIEW
Connect to a Remote Server
New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB): 15.0
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel
vmware


VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Ubuntu 64-bit (3)
Ubuntu 64-bit (4)
Home
My Computer
Ubuntu 64-bit (2)
Ubuntu 64-bit (3)
Clone of Ubuntu 64-bit (4)
WORKSTATION 20H2™
TECH PREVIEW
Connect to a Remote Server
New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine. Then you can install Ubuntu 64-bit.
The virtual machine will be created with the following settings:
Name: Ubuntu 64-bit (5)
Location: C:\Users\prana\OneDrive\Documents\Virtual Machines\...
Version: Workstation Beta
Operating System: Ubuntu 64-bit
Hard Disk: 15 GB
Memory: 2048 MB
Network Adapter: NAT
Other Devices: CD/DVD, USB Controller, Printer, Sound Card
Customize Hardware...
< Back
Finish
Cancel
vmware


Ubuntu 64-bit (5) - VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Ubuntu 64-bit (3)
Ubuntu 64-bit (4)
Ubuntu 64-bit (5)
Home
My Computer
Ubuntu 64-bit (2)
Ubuntu 64-bit (3)
Clone of Ubuntu 64-bit (4)
Ubuntu 64-bit (5)
Ubuntu 64-bit (5)
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 2 GB
Processors 1
Hard Disk (SCSI) 15 GB
CD/DVD (SATA) Auto detect
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Printer Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\prana\OneDrive\Documents\Virtual Machines\Ubuntu 64-bit (5)\Ubuntu 64-bit (5).vmx
Hardware compatibility: Workstation Beta virtual machine
Primary IP address: Network information is not available



Result:The virtual machine was successfully created using VMware with 1 CPU, 2 GB RAM and 15 GB storage.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

AIM: To create a clone of an existing virtual machine and test the cloned VM by restoring its previous version.

PROCEDURE:

STEP 1: Go to VM and goto manage and clock clone.

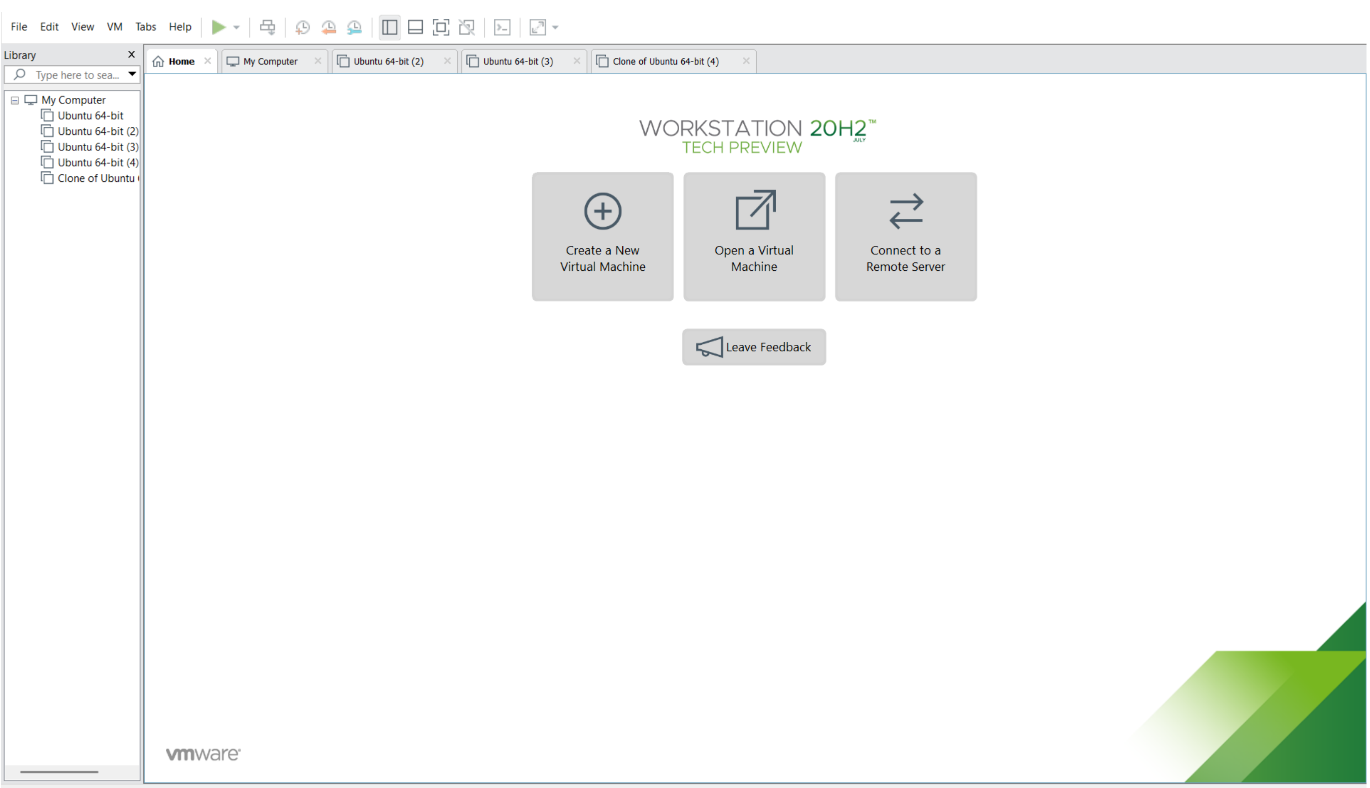
STEP 2: Click Clone

STEP 3: Select the full clone

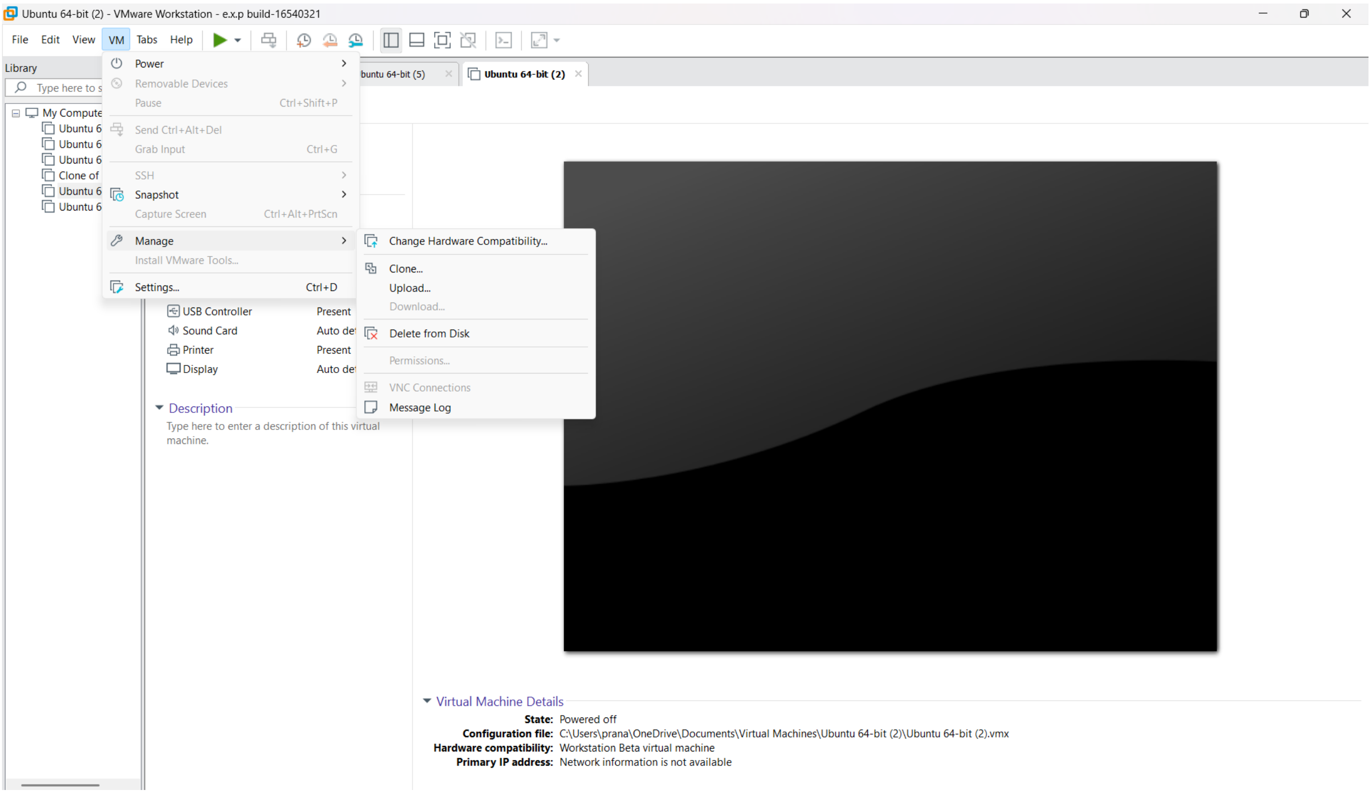
STEP 4: After clone again or VM is opened

ALGORITHM:

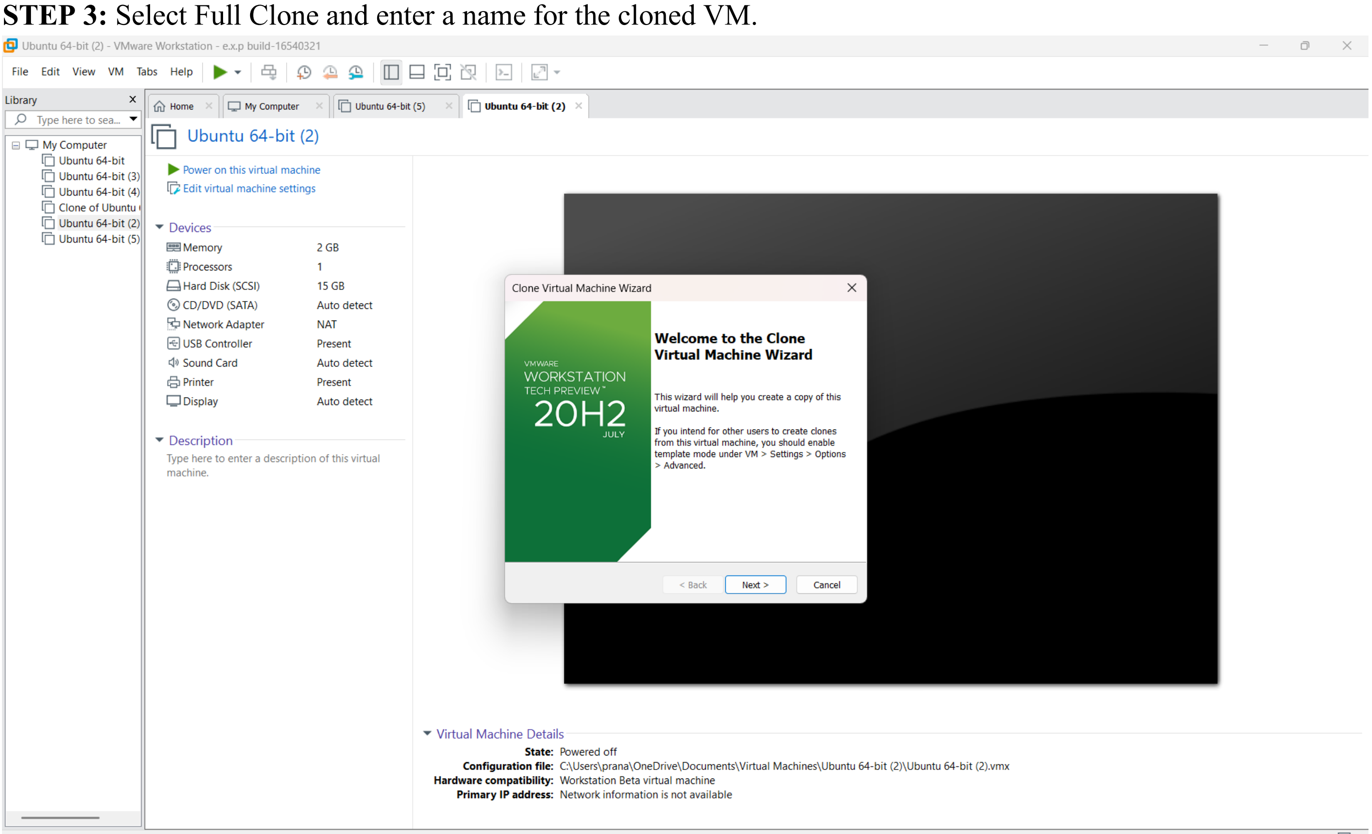
STEP 1: DOWLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2HYPERVISOR



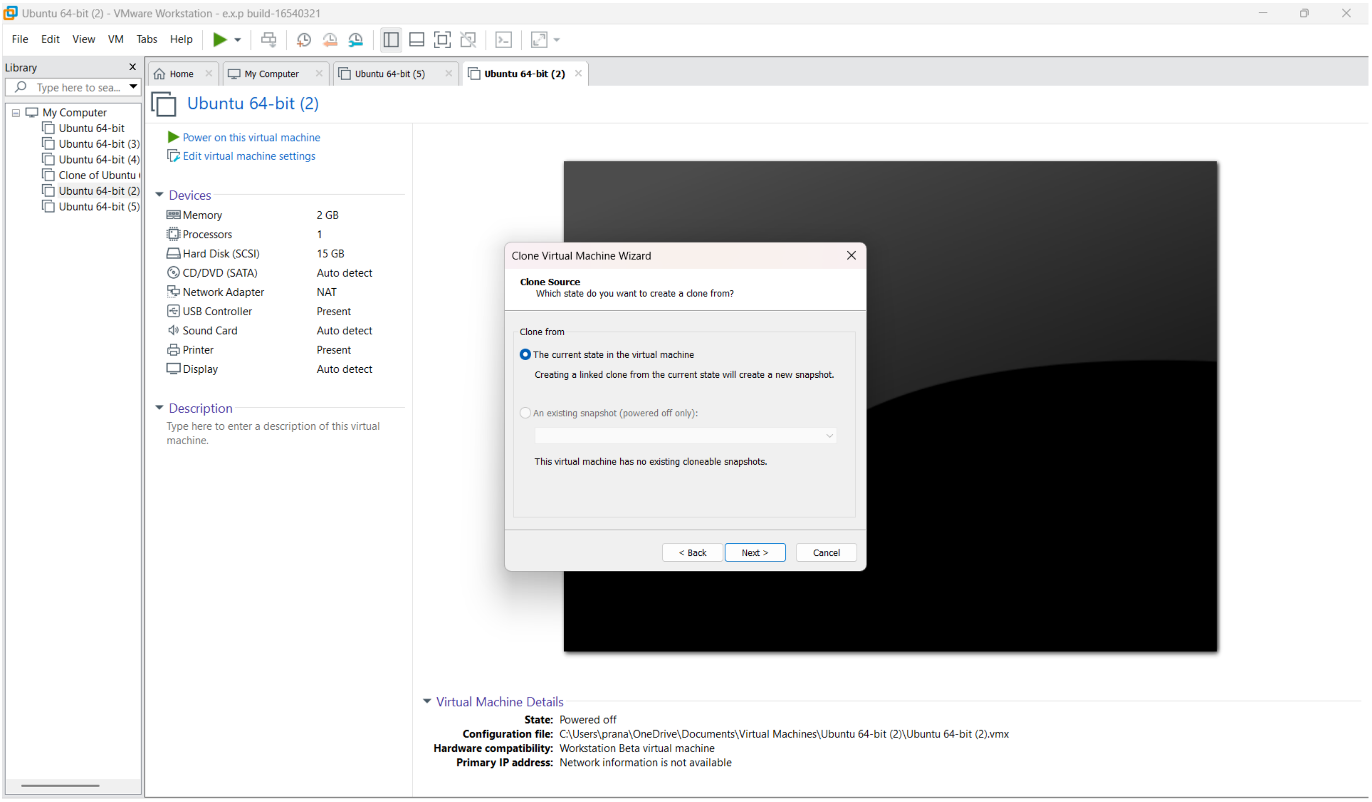
STEP 2: Select Manage → Clone to start the cloning process.

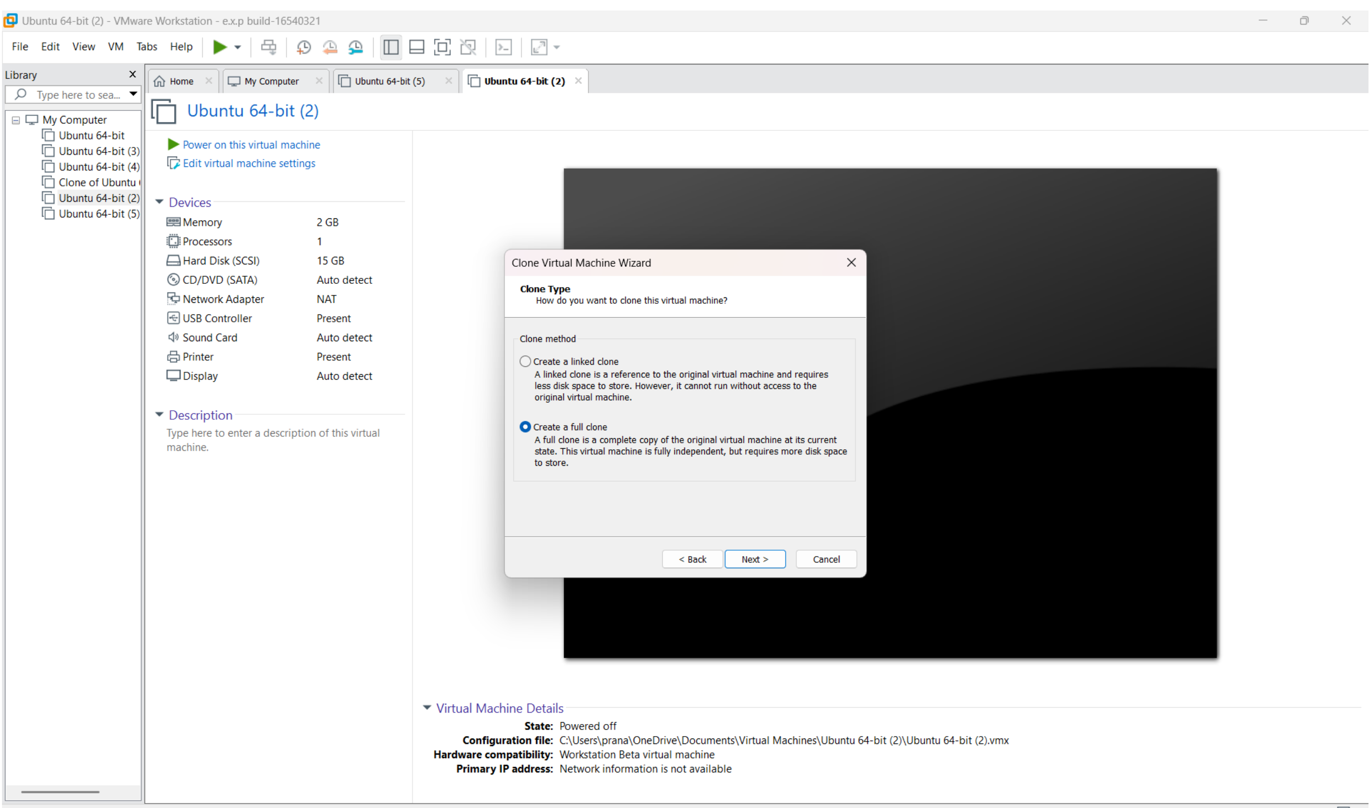


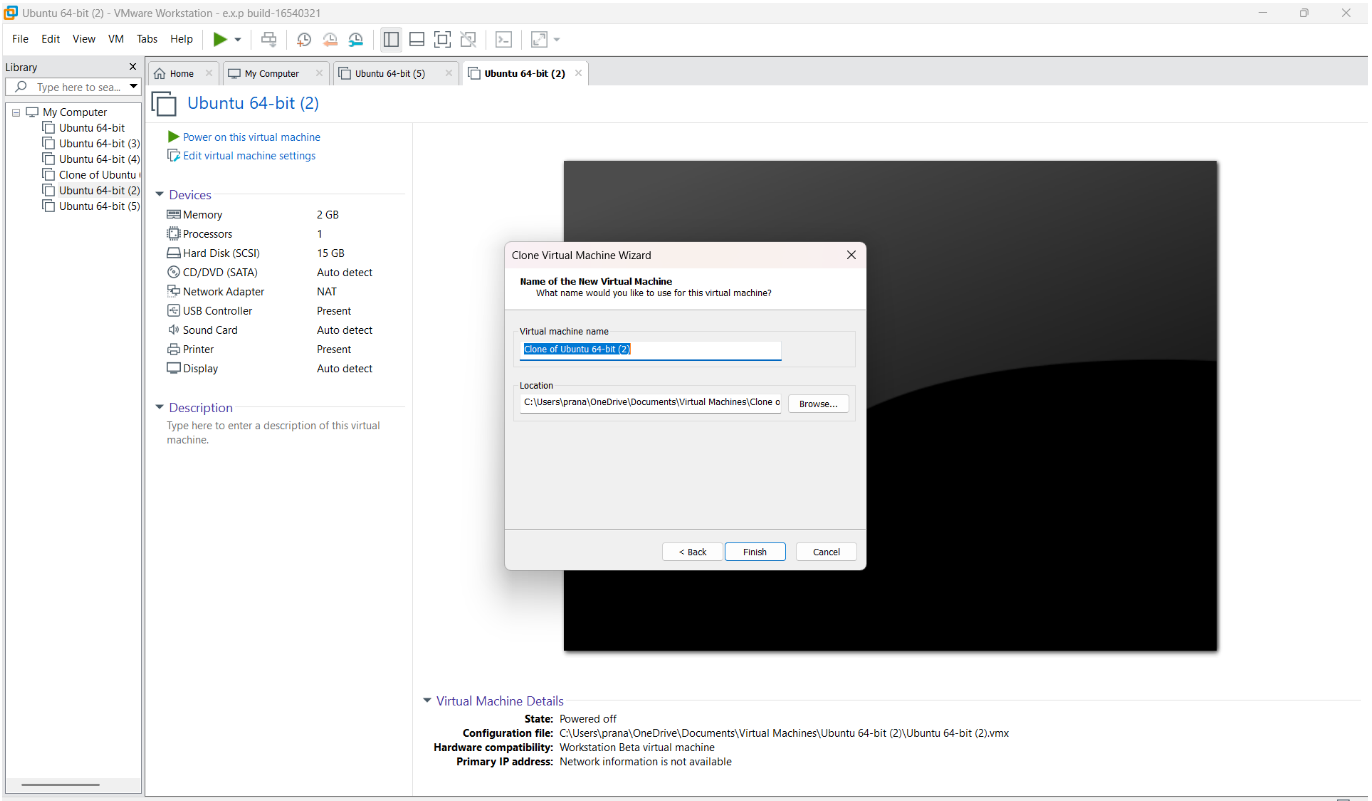
STEP 3: Select Full Clone and enter a name for the cloned VM.

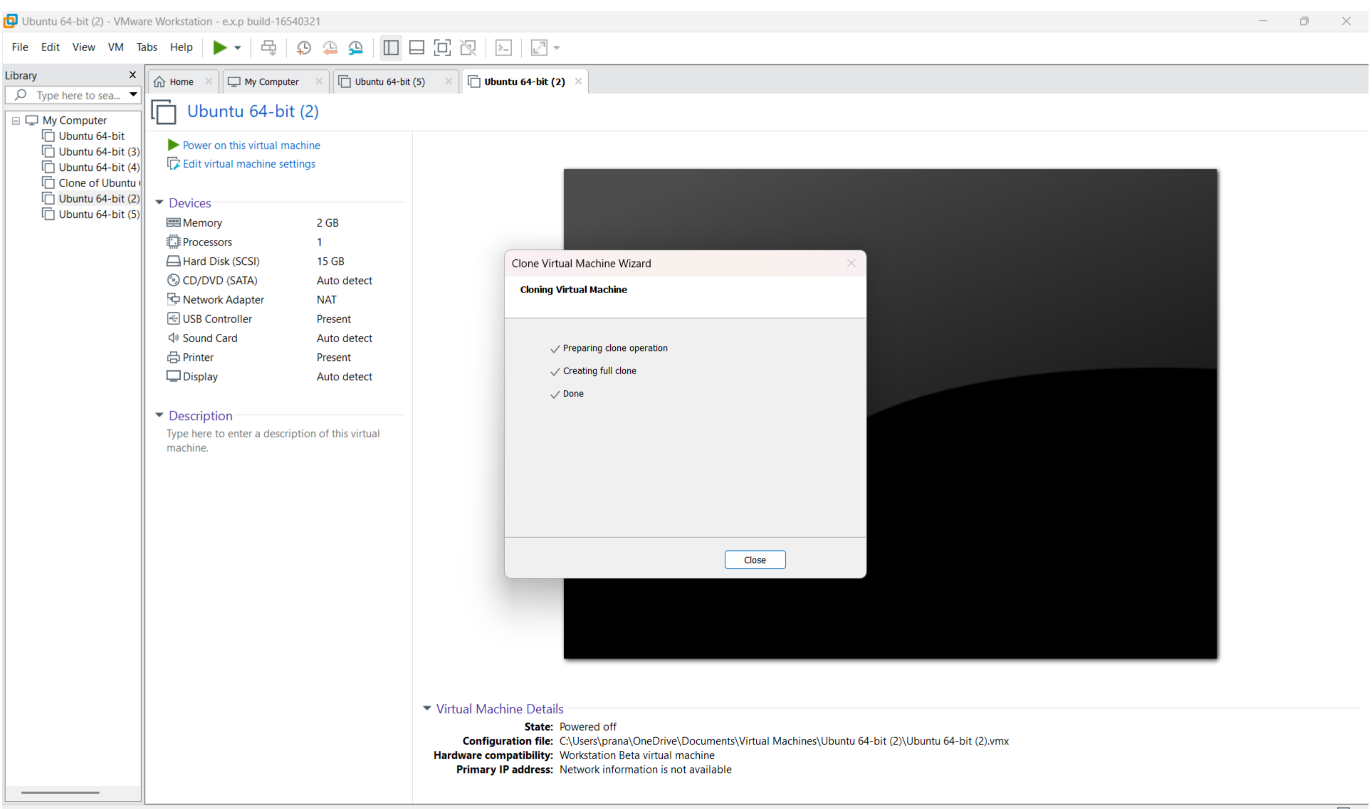


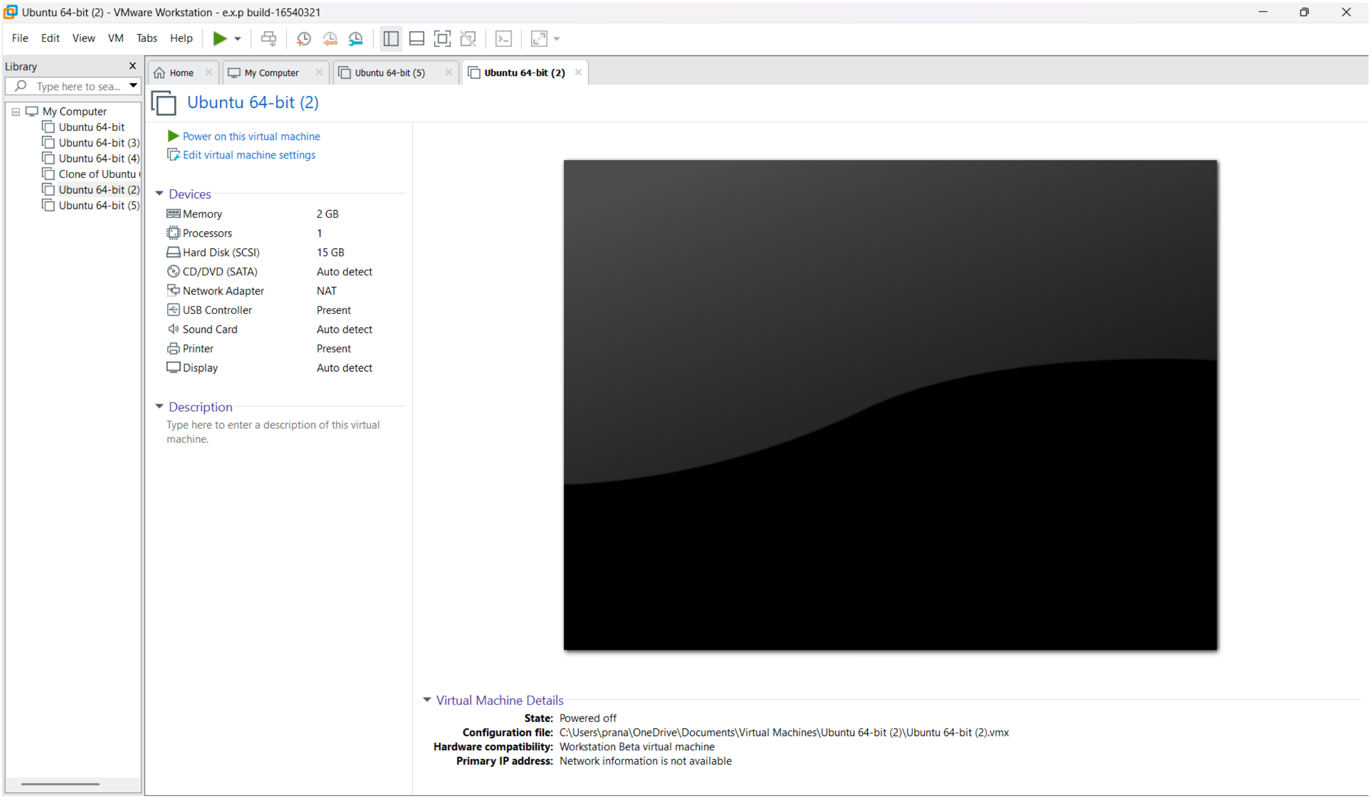
STEP 4: Complete the cloning process and start the cloned VM.






Ubuntu 64-bit (2) - VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (3)
Ubuntu 64-bit (4)
Clone of Ubuntu
Ubuntu 64-bit (2)
Ubuntu 64-bit (5)
Home
My Computer
Ubuntu 64-bit (5)
Ubuntu 64-bit (2)
Ubuntu 64-bit (2)
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 2 GB
Processors 1
Hard Disk (SCSI) 15 GB
CD/DVD (SATA) Auto detect
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Printer Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Clone Virtual Machine Wizard
Name of the New Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name
Clone of Ubuntu 64-bit (2)
Location
C:\Users\prana\OneDrive\Documents\Virtual Machines\Clone o
Browse...
< Back
Finish
Cancel
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\prana\OneDrive\Documents\Virtual Machines\Ubuntu 64-bit (2)\Ubuntu 64-bit (2).vmx
Hardware compatibility: Workstation Beta virtual machine
Primary IP address: Network information is not available


Ubuntu 64-bit (2) - VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (3)
Ubuntu 64-bit (4)
Clone of Ubuntu
Ubuntu 64-bit (2)
Ubuntu 64-bit (5)
Home
My Computer
Ubuntu 64-bit (5)
Ubuntu 64-bit (2)
Ubuntu 64-bit (2)
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 2 GB
Processors 1
Hard Disk (SCSI) 15 GB
CD/DVD (SATA) Auto detect
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Printer Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Clone Virtual Machine Wizard
Cloning Virtual Machine
Preparing clone operation
Creating full clone
Done
Close
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\prana\OneDrive\Documents\Virtual Machines\Ubuntu 64-bit (2)\Ubuntu 64-bit (2).vmx
Hardware compatibility: Workstation Beta virtual machine
Primary IP address: Network information is not available



Result: The VM was successfully cloned and tested, and its previous saved state was successfully restored.